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Project Title: Phase-Bounded Gait Event Detection and prediction during STW transition using ML

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ABSTRACT

Human locomotion is a complex, structured process that is controlled by a series of biomechanical events. Gait event detection and walking phase segmentation are critical for several applications in rehabilitation engineering, assistive robotics, prosthetics, and clinical gait analysis. Of all human walking activities, the Sit-to-Walk transition is a complex movement, as it involves a rapid transition between a quasi-static posture in sitting and a dynamic walking state, with non-periodic transient phases.

This graduation project proposes a complete, reproducible framework for phase-bound gait event detection and walking phase segmentation using wearable sensors, with a focus on IMU-EMG analysis. Surface electromyography (sEMG) signals from eight muscles in the lower limbs, as well as inertial measurement unit (IMU) data from multiple body segments, are processed, with special emphasis placed on foot-worn IMU sensors. The proposed approach for phase-bound gait event detection and walking phase segmentation starts with robust data handling, including synchronization of multi-rate EMG signals, elimination of non-finite value occurrences, and subsequent application of appropriate filtering techniques. Heel strike and toe-off are automatically identified using the X-axis gyroscope signal from the foot IMU, which allows for the definition of individual gait cycles.

Each gait cycle has four phases: Loading Response (LR), Mid-Stance (MS), Terminal Stance (TS), and Swing (SW). Time-domain signal features are extracted within each phase window for both EMG and IMU data: Root Mean Square (RMS), Integrated EMG (IEMG), Mean Absolute Value (MAV), Waveform Length (WL), Zero Crossings (ZC), Slope Sign Changes (SSC), Total Power (TP), and Peak Frequency (PF). These are then prepared for a machine learning classification task.

The system employs Random Forest for gait phase classification and Long Short-Term Memory (LSTM) networks for next-event prediction, evaluated using subject-wise cross-subject validation. The proposed framework demonstrates high classification accuracy and strong temporal prediction performance under real-world wearable sensor conditions, providing a robust and scalable foundation for future rehabilitation, assistive robotics, and intelligent gait monitoring applications.



Figure 1: Overview of the system

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Chapter 1 – Introduction

1.1 Background

Human locomotion is a fundamental function of daily life, and it serves as a critical indicator of neuromuscular function and motor control. Walking is a nonrandom motion, and it consists of a predictable and repetitive sequence of events and phases. It represents a complex interaction of the musculoskeletal system, nervous system, and environment. It plays a critical role in the accurate understanding and detection of gait events and phases in many application domains, including clinical gait assessment, rehabilitation engineering, assistive robotics, prosthetics, exoskeleton control, fall risk monitoring, and sports biomechanics.

Traditionally, gait analysis has been carried out in a laboratory setting using optical motion capture systems, force plates, and instrumented treadmills. These systems offer accurate and precise data regarding the gait of a person, and it becomes easier to identify the events of gait, including heel strike and toe-off. It also becomes easier to segment the gait cycle into different phases, including stance and swing phases. However, these systems have several disadvantages, including being expensive and only capable of providing accurate data in a laboratory setting.

The recent technological improvements in wearable sensing systems have made it feasible to change the paradigm from stationary gait analysis to portable and unobtrusive gait analysis. Inertial Measurement

Units (IMUs), which consist of accelerometers and gyroscopes, have been used to measure the motion of body segments. Muscle activation patterns, which describe the neuromuscular control strategy used in movement, have been captured using surface Electromyography (sEMG) sensors. The integration of IMU and EMG sensing systems has been found to be a powerful tool for describing events in kinematics and muscle activations associated with movement.

Of all the daily locomotion activities, Sit-to-Walk (STW) transition is a particularly difficult movement. It consists of a quick transition from a quasi-static position to a dynamic walking position. Unlike steady-state walking, the Sit-to-Walk transition comprises several transient events, including trunk flexion, seat-off, weight transfer, and gait initiation. These events have unique non-periodic kinematics and muscle activation patterns. As a result, the detection and segmentation of Sit-to-Walk transition require sophisticated signal processing and classification techniques compared to those used for steady-state walking.

1.2 Motivation

This project has direct implications for real-world applications. For example, in rehabilitation and assistive technology, exoskeletons and prosthetic devices must coordinate their assistance with the user's intended gait. Providing assistance too early or too late for phases such as loading response or swing can negatively impact the effectiveness of the device. Similarly, abnormal gait patterns and delayed transitions detected during STW can offer important cues for neurological and musculoskeletal disorders.

Recently, machine learning (ML) methods have demonstrated promise for extracting meaning from complex and multidimensional sensor information. When paired with wearable sensors, ML methods can learn discriminative patterns from sensor information, which might be challenging using traditional rule-based methods. However, the effectiveness of ML methods is heavily dependent upon the quality of the input information and corresponding labels. Sensor information is inherently noisy and contains various types of artifacts. Sensor sampling frequencies are often inconsistent across channels, and ML methods can easily get confused if not appropriately preprocessed and segmented.

The reason for undertaking this project is the need for a systematic and repeatable pipeline that closes the gap between raw wearable sensor data and machine learning-based gait phase classification. In contrast to other projects that concentrate on model complexity, this project stresses the relevance of early-stage processing steps such as data cleaning, synchronization, filtering, event detection, and phase-bound segmentation. By doing so, the dataset is made suitable for proper ML training and testing.

1.3 Problem Statement

Despite the presence of wearable IMU and EMG sensors, the following challenges make it difficult to automatically detect gait phases and analyze STW:

1. Noise and Artifacts in Sensors:

EMG signals are prone to artifacts caused by motion, electrode movement, and powerline noise. IMU signals can have high-frequency noise and low-frequency drift. These problems can mask useful physiological data if not handled appropriately.

2. Multi-Rate Data Acquisition:

In real-world datasets, different sensors or channels can be acquired at different rates. In the given dataset, some EMG channels are acquired at 1259 Hz, while others are acquired at 1778 Hz. Direct comparison or simultaneous feature extraction without synchronization results in misalignment and incorrect analysis.

3. Non-Finite and Invalid Samples:

Real-world datasets can have missing values, NaNs, infinite values, or trailing zeros to mark the end of a recording. These values can cause mathematical failures during filtering and incorrect calculations of statistical features.

4. Event Detection without Ground Truth:

Without the use of force plates or motion capture systems, gait events such as heel strike and toe-off need to be detected solely from wearable sensors. Developing reliable and consistent event detection algorithms is a major challenge.

5. Consistent Phase Definition:

In the development of the machine learning dataset, the continuous data must be divided into consistent and physiologically meaningful phase windows. Inexact and inconsistent definitions directly affect the quality of the label.

In view of the aforementioned issues, the primary problem that the project aims to resolve is the development of an end-to-end processing pipeline that can convert the raw wearable sensor data into a clean, synchronized, and labeled dataset for the classification of gait phases and STW transition analysis.

1.4 Objectives of the Project

The major objectives of this graduation project can be summarized as follows:

1. Importing and parsing wearable sensor data recorded in CSV file formats, recognizing relevant EMG and IMU channels based on their physical placement.

2. Cleaning raw sensor data by handling non-finite values, eliminating invalid data, and truncating data at appropriate termination points.

3. Synchronizing multi-rate EMG channels by resampling them at a common reference sampling frequency.

4. Applying appropriate filtering techniques for EMG and IMU data, such as band-pass filtering, notch filtering, rectification, smoothing, etc.

5. Detecting important gait events such as heel strike and toe-off using foot-mounted IMU gyroscope data.

6. Segmenting each walking cycle into standard phases of walking, i.e., LR, MS, TS, and SW.

7. Extracting important time-domain features from EMG and IMU data within each walking phase.

8. Creating a machine learning dataset for evaluating walking phase classification.

9. Evaluating basic machine learning models for their potential use in STW transition detection/prediction.

1.5 Scope of the Project

This project is specifically aimed at offline analysis and model development using pre-recorded wearable sensor data. The dataset consists of walking trials at varying speeds, such as slow, normal, and fast. However, the major classification task in this case is related to recognizing a four-phase gait. Although the overall idea is related to recognizing transitions related to STW, the current project has been developed based on steady-state walking phases, which can be later extended to include sub-phases related to STW.

Although the current project does not involve real-time embedded system development or clinical validation for large populations, emphasis has been given to methodological correctness, replicability, and documentation of each step in the process.

1.6 Methodology Overview

1. Data Import and Channel Selection:

The raw CSV data is imported, and the corresponding EMG and IMU channels are selected based on known column mappings.

2. Data Cleaning and Synchronization:

Non-finite data is processed, trailing zeros are removed, and all EMG channels are resampled to a common sampling rate.

3. Signal Conditioning:

The EMG signals are band-pass and notch filtered, rectified, and optionally smoothed. The IMU gyroscope signals are low-pass or band-pass filtered to highlight gait cycle dynamics.

4. Event Detection:

Heel strike and toe-off events are identified from the foot IMU gyroscope signal based on peak and timing thresholds.

5. Phase Segmentation:

Each stride is divided into LR, MS, TS, and SW phases based on event markers and stride-relative timing rules.

6. Feature Extraction:

Time-domain features are extracted for each EMG and IMU channel within the phase windows.

7. Dataset Construction and ML Training:

Feature vectors and labels are combined into a machine learning dataset and used to train and test baseline classifiers.

1.7 Contributions of This Work

The key contributions of this project are:

- A reproducible MATLAB code for processing multi-rate EMG and IMU data.
- Effective handling of non-finite values for proper filtering and feature extraction.
- Automatic gait event detection using a foot-worn IMU gyroscope signal.
- Uniform segmentation of walking cycles into four gait phases based on physiological characteristics.

- Building a machine learning dataset for gait phase classification and analysis for STW.

1.8 Structure of the Report

The rest of the report is organized as follows

Chapter 2: provides the theoretical foundation that is required for understanding human gait, Sit-to-Walk (STW) transitions, wearable sensors using IMUs and EMG, signal characteristics, filtering, and feature-based machine learning.

Chapter 3: presents the related work, focusing on the fields of gait event detection, gait phase classification, Sit-to-Walk, and machine learning on wearable sensors, including the advantages and disadvantages of the related work.

Chapter 4: covers the dataset and experimental setup, including sensor configuration, placement of sensors, and the structure of the recorded data.

Chapter 5: describes the complete signal processing flow used on the raw sensor data, including cleaning, handling of non-finite numbers, synchronization of multi-rate signals, and filtering of EMG and IMU signals.

Chapter 6: covers the strategy used for gait event detection and phase segmentation, followed by feature extraction of EMG and IMU signals and the generation of the labeled dataset for machine learning.

Chapter 7: discusses the machine learning used for gait phase classification, including training and validation.

Chapter 8: presents the details of the results obtained, including the confusion matrix of the performance, interpretation of the segmentation results, limitations of the suggested approaches, and a comparison of the approaches used.

Finally, Chapter 9: concludes the entire report with a summary of the main contributions of the project, including the direction for future research.

Appendices:

The appendices present supplementary materials, including diagrams of sensor locations, column mappings, MATLAB code snippets, plots, and tables.

References: List of references for the report

Chapter 2 – Background and Theoretical Foundations

2.1 Overview of Human Gait

Human gait is a repetitive and coordinated movement pattern that allows for the forward advancement of the body, along with the maintenance of balance and stability. In terms of biomechanics, gait is the outcome of intricate relationships between the musculoskeletal, neuromuscular, and external forces, including gravity and ground reaction forces.
To understand gait, it is essential to analyze the kinematic and neuromuscular aspects of the gait pattern.

A gait cycle is traditionally defined as the time period between the occurrence of two consecutive heel strikes by the same foot. In the process, the limbs switch between the stance and swing phases, where the limbs are either in contact with the ground or moving forward with no ground contact, respectively. Although the process of walking is considered continuous, it can be divided into discrete phases, which can be considered unique biomechanical functions, including shock absorption, body support, propulsion, and advancement of the limbs.

The process of gait analysis is intended to identify the phases and the events with high temporal resolution. The identification process is vital, especially for applications where the process is considered essential, including robotics, prosthetics, and diagnosis.

2.2 Gait Events and Phase Definitions

2.2.1 Gait Events

Gait events are discrete temporal markers that indicate transitions between phases of the gait cycle. These events are a prerequisite for segmenting continuous locomotion data into physiologically meaningful units and defining temporal bounds of gait phases. In wearable-sensor-based gait analysis, events are typically inferred from kinematic signals, with a special interest in the angular velocity measured by a foot-mounted IMU.

The two basic gait events used to define a gait cycle are: **Heel-Strike (HS)**, and **Toe-Off (TO)**. These events delimit the stance and swing phases and can be detected robustly across different walking conditions.

Heel-Strike (HS)

Heel-strike is the instant when the heel of the foot touches the ground. In the case of a foot-mounted IMU, this event corresponds to a well-marked extremum in the angular velocity signal due to the rapid deceleration of the foot. In this work, heel-strike is detected from the gyroscope X-axis signal, denoted as $\omega(t)$, by the identification of a local minimum within a predefined temporal window. The angular velocity at heel-strike can be defined as:

$$\omega_{HS} = min(\omega(t)), t \in [t_{MSw}, t_{TO\text{-}next}] \quad \ldots (1)$$

where t_{MSw} denotes the time of mid-swing and $t_{TO\text{-}next}$ denotes the next toe-off event. This formulation ensures that heel-strike is detected after the swing phase and before the subsequent stance phase begins.

Toe-Off (TO)

Toe-off marks the instant when the toes lose contact with the ground, signifying the transition from stance to swing. In foot gyroscope signals, toe-off is also characterized by a distinct angular velocity extremum, typically occurring before mid-swing.

Toe-off is detected as:

$$\omega_{TO} = min(\omega(t)), t \in [t_{HS\text{-}last}, t_{\mathrm{MS}w}] \quad \ldots (2)$$

with the additional constraint:

$$\omega_{TO} < \omega_{HS}$$

This inequality ensures that the detected toe-off event is distinguishable from the subsequent heel-strike extremum. The temporal window is bounded by the previous heel-strike and the mid-swing instant.

In addition to HS and TO, secondary gait events are often defined to refine stance-phase segmentation.

Foot Flat (FF)

Foot flat occurs when the entire plantar surface of the foot is in contact with the ground, typically following heel-strike. During this period, foot rotation is minimal, and the angular velocity signal remains close to zero.
Foot flat is defined as the interval in which:

$$|\omega(t)| < \varepsilon, |\frac{d\omega(t)}{dt}| \approx 0, t \in [t_{HS}, t_{TO}] \quad \ldots (3)$$

where ε is a small threshold accounting for sensor noise. This condition reflects a quasi-static foot posture during weight acceptance.

Heel Off (HO)

Heel off corresponds to the instant when the heel lifts from the ground while the forefoot remains in contact. This event marks the transition from mid-stance toward terminal stance and precedes toe-off.

Heel off is characterized by:

$$\omega(t_{HO}) < 0, \frac{d\omega(t)}{dt}\Big|_{t=t_{HO}} = t_{HO} > 0, t \in [t_{FF}, t_{TO}] \quad \ldots (4)$$

This definition captures the onset of forward foot rotation and increasing angular velocity associated with propulsion.

While HS and TO define the fundamental structure of the gait cycle, FF and HO provide additional insight into stance sub-phases and improve biomechanical interpretation when such refinement is required.

Mid-Swing (MSw)

Mid-swing is a discrete event occurring during the swing phase, approximately when the swinging foot reaches its maximum forward angular velocity. Biomechanically, this instant corresponds to the foot passing beneath the body while advancing toward the next heel contact. In foot gyroscope signals, mid-swing is typically associated with a prominent positive peak.

Mid-swing is defined as:

$$\omega_{MSw} = \max(\omega(t)), t \in [t_{TO}, t_{HS}] \quad \ldots (5)$$

where $\omega(t)$denotes the gyroscope Z-axis angular velocity of the foot IMU. The mid-swing event is particularly useful for bounding the temporal search windows used to detect heel-strike and toe-off events, thereby improving robustness in noisy signals.

Mid-Stance (MSt)

Mid-stance is a reference event occurring during the stance phase when the body's center of mass is approximately aligned over the supporting foot. At this moment, rotational motion of the foot is minimal, and the angular velocity measured by the gyroscope approaches zero.

Mid-stance is defined as:

$$\omega_{MSt} = min(\omega(t)) \cong 0, t \in [t_{HS}, t_{TO}] \quad \ldots (6)$$

This condition reflects a quasi-static support phase and provides a useful internal reference for identifying secondary events such as heel-off and for validating stance-phase segmentation.

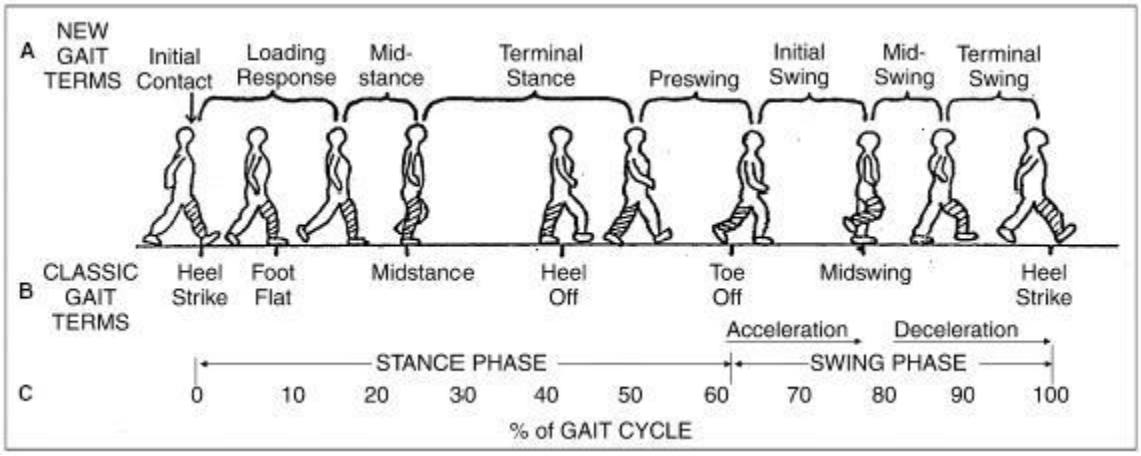


Figure 2: Gait phases & Events

2.2.2 Gait Phase Definitions

Based on the detected gait events, the gait cycle is divided into phases that reflect distinct biomechanical functions. In this project, both stance and swing phases are further characterized using gyroscope-based definitions.

1. Loading Response (LR):
Begins at heel-strike and ends when the body weight is fully accepted by the stance leg. This phase is characterized by shock absorption and rapid adjustments in muscle activity to stabilize the joints.

2. Mid-Stance (MS):
Occurs when the body's center of mass passes over the supporting foot. During mid-stance, the limb provides support and balance while the contralateral limb is in swing.

3. Terminal Stance (TS):
Represents the late stance period, where the heel lifts off the ground and propulsion is generated in preparation for toe-off.

4. Swing (SW):
Begins at toe-off and continues until the next heel-strike. The limb advances forward and prepares for the next cycle.

This four-phase representation is widely used in wearable-sensor-based gait analysis, as it provides sufficient temporal resolution for classification while remaining robust to sensor noise and timing uncertainty.

2.3 Sit-to-Walk (STW) Transition

2.3.1 Definition of STW

Sit-to-Walk (STW) is a compound movement task, and it is the combination of sit-to-stand and gait initiation movement tasks in a continuous manner. In the steady-state walking movement, sit-to-walk occurs without any rest between the sit-to-stand and gait initiation movement tasks.

2.3.2 Biomechanical Characteristics

STW includes several sub-components:

- Trunk flexion to shift the center of mass forward
- Seat-off event
- Weight transfer to the stance limb
- Initiation of the first step

These components overlap in time and vary significantly between individuals, making STW segmentation more challenging than steady walking.

2.3.3 Relevance to This Project

Although this project primarily focuses on walking-phase segmentation, STW is a key motivation. By establishing a robust method for detecting gait events and phases during walking. The system detected the duration of the STW through every trial (when it starts and when it ends), so that can later be extended to STW-specific sub-phase detection and prediction.

2.4 Wearable Sensors for Gait Analysis

2.4.1 Inertial Measurement Units (IMUs)

IMUs are usually composed of:

- Accelerometers (linear acceleration)
- Gyroscopes (angular velocity)
- Magnetometers (orientation reference, optional)

IMUs are widely used in gait analysis because they are portable and inexpensive and can measure movement in a free environment.

Foot-Mounted IMUs

Among all sensor placements, foot-mounted IMUs are particularly effective for gait event detection. The angular velocity around the mediolateral axis (often represented by the Z-axis, depending on sensor orientation) exhibits characteristic patterns during heel-strike and toe-off, making it well suited for event

detection. But on the experiment the Rotation was happened on the X-axis not on the Z-axis.



Figure 3: Foot IMU

2.4.2 Surface Electromyography (sEMG)

Surface EMG measures electrical activity generated by muscle fibers during contraction. sEMG provides insight into the timing and intensity of muscle activation and is therefore essential for understanding neuromuscular control.

In gait analysis, EMG signals are commonly recorded from lower-limb muscles such as the tibialis anterior, gastrocnemius, quadriceps, and hamstrings. Each muscle exhibits phase-dependent activation patterns, which can be exploited for gait phase classification.



Figure 4: sEMG Sensors

2.4.3 Complementarity of IMU and EMG

IMU and EMG signals provide complementary information:

- IMUs capture **kinematic events** (movement and orientation)
- EMG captures **neuromuscular intent and activation**

Combining both modalities improves robustness, particularly in challenging transitions such as STW.

2.5 Signal Characteristics and Challenges

2.5.1 EMG Signal Characteristics

EMG signals are stochastic and broadband in nature. Typical characteristics include:

- Frequency content primarily between 20–450 Hz
- Sensitivity to motion artifacts at low frequencies
- Susceptibility to powerline interference (50/60 Hz)

Raw EMG cannot be directly used for feature extraction without appropriate preprocessing.

2.5.2 IMU Signal Characteristics

IMU gyroscope signals are smoother than EMG but can contain:

- High-frequency noise
- Low-frequency drift
- Sensor bias

Filtering is therefore required to isolate gait-related motion patterns.

2.5.3 Multi-Rate Sampling

In practical datasets, different sensors or channels may operate at different sampling frequencies. In the dataset used in this project, EMG channels are recorded at both 1259 Hz and 1778 Hz. Without resampling, direct comparison or joint feature extraction across channels would be invalid.

2.6 Filtering Techniques in Gait Analysis

2.6.1 Band-Pass Filtering

Band-pass filtering is commonly applied to EMG signals to remove:

- Low-frequency motion artifacts
- High-frequency noise

A typical EMG band-pass range is 20–450 Hz.

2.6.2 Notch Filtering

Notch filters are used to suppress powerline interference, commonly at 50 Hz in many regions.

2.6.3 Low-Pass Filtering

Low-pass filtering is often applied to:

- IMU signals to emphasize gait dynamics
- Rectified EMG envelopes to obtain smoothed activation profiles

2.6.4 Alternative Filters

Other filters used in wearable-sensor processing include:

- **Moving average filters:** simple smoothing
- **Savitzky–Golay filters:** preserve waveform shape while smoothing
- **Gaussian filters:** smooth signals with minimal ringing
- **Interpolation filters:** reconstruct missing or irregular samples

The choice of filter depends on signal characteristics and application requirements.

2.7 Feature-Based Machine Learning for Gait Analysis

2.7.1 Time-Domain Features

Time-domain features are widely used due to their low computational cost and interpretability. Common features include:

- Root Mean Square (RMS)
- Integrated Electromyography (IEMG)
- Mean Absolute Value (MAV)
- Waveform Length (WL)
- Zero Crossings (ZC)
- Slope Sign Changes (SSC)

These features capture amplitude, complexity, and frequency-related information indirectly.

2.7.2 Frequency-Domain Features

Frequency-domain features are used to describe the spectral characteristics of EMG and IMU signals by analyzing the distribution of signal energy across frequencies. These features provide complementary information to time-domain features and are particularly useful for capturing dominant activation frequencies and overall signal energy within a gait phase.

Common frequency-domain features include:

- Peak Frequency (PF)
- Total Power (TP)

These features are typically computed using power spectral density estimation methods and are extracted from phase-bounded signal windows to ensure biomechanical relevance.

2.7.3 Phase-Bounded Feature Extraction

Rather than extracting features from fixed-length sliding windows, this project uses **phase-bounded windows**, where features are computed separately for LR, MS, TS, and SW. This approach ensures that each feature vector corresponds to a physiologically meaningful movement phase.

2.7.4 Advantages of Feature-Based Models

For moderate dataset sizes, classical feature-based models such as Random Forests and Support Vector Machines:

- require less data than deep learning,
- provide better interpretability,
- are easier to debug and validate.

2.8 Summary

This chapter provided the theoretical foundation required to understand gait phase segmentation and STW analysis using wearable sensors. Key concepts related to gait biomechanics, STW transitions, IMU and EMG sensing, signal characteristics, filtering methods, and feature-based machine learning were discussed. These concepts motivate the design choices made in subsequent chapters.

The next chapter presents a detailed review of related work in gait analysis, STW detection, and wearable-sensor-based machine learning approaches.

Chapter 3 – Related Work

3.1 Overview

Research on gait analysis has a long history in biomechanics, rehabilitation engineering, and robotics. With the increasing availability of wearable sensors and advances in machine learning, recent work has shifted from laboratory-based systems toward portable, data-driven approaches. This chapter reviews existing literature relevant to this project, focusing on gait event detection, gait phase segmentation, Sit-to-Walk (STW) analysis, wearable IMU and EMG sensing, and machine learning methods applied to locomotion data.

The review highlights commonly used methodologies, sensor configurations, and algorithmic strategies, and it identifies limitations that motivate the approach adopted in this project.

3.2 Traditional Gait Analysis Methods

Early gait analysis studies relied primarily on optical motion capture systems combined with force plates. These systems allow precise measurement of joint kinematics and ground reaction forces, enabling accurate detection of gait events such as heel-strike and toe-off. Force plate data are often considered the gold standard for gait event detection due to their high temporal accuracy.

However, traditional systems suffer from several drawbacks:

- High cost and complex setup
- Requirement for controlled laboratory environments
- Limited capture volume
- Inability to support long-term or real-world monitoring

As a result, their applicability outside specialized laboratories is limited, motivating the exploration of wearable alternatives.

3.3 IMU-Based Gait Event Detection

3.3.1 Foot-Mounted IMU Approaches

A large body of literature demonstrates that foot-mounted IMUs are particularly effective for gait event detection. Angular velocity signals measured by foot gyroscopes exhibit characteristic patterns during heel-strike, toe-off, and swing. Many studies detect heel-strike as a local extremum or zero-crossing in the gyroscope signal and toe-off as a distinct angular velocity feature occurring later in the stance phase.

Common strategies include:

- peak detection with temporal constraints
- zero-crossing analysis
- threshold-based detection on angular velocity magnitude

Foot-mounted IMUs offer advantages such as robustness to upper-body motion and strong correlation with foot-ground interaction events.

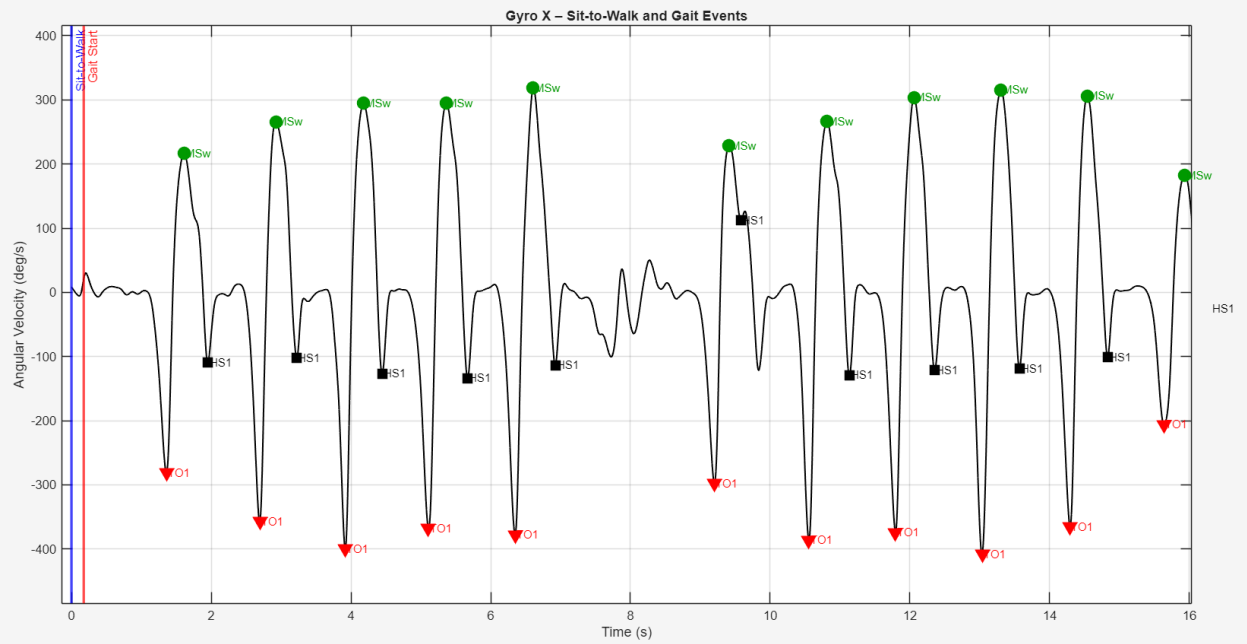


Figure 5: Foot IMU - Gyro X Signal – Gait Events Detection

3.3.2 Multi-IMU Configurations

Some studies employ multiple IMUs placed on the shank, thigh, pelvis, or trunk to improve robustness and enable full-body kinematic reconstruction. While multi-sensor configurations provide richer information, they increase system complexity, computational cost, and susceptibility to sensor misalignment.

In contrast, single foot-mounted IMU approaches strike a balance between accuracy and practicality, making them well suited for wearable gait phase detection systems.

3.4 EMG-Based Gait Phase Analysis

Surface electromyography (sEMG) has been widely used to analyze neuromuscular activity during gait. Different lower-limb muscles exhibit characteristic activation patterns depending on the gait phase. For example:

- Tibialis Anterior is active during swing and initial contact
- Gastrocnemius muscles are active during late stance and push-off
- Quadriceps muscles contribute during loading response and early stance

Several studies use EMG features to classify gait phases directly or to supplement kinematic data. Commonly extracted features include RMS, IEMG, MAV, WL, ZC, and SSC, computed over sliding or phase-based windows.

However, EMG-only approaches face challenges:

- high inter-subject variability
- sensitivity to electrode placement
- susceptibility to noise and motion artifacts

As a result, EMG is often combined with IMU data to improve robustness.

3.5 Combined IMU–EMG Approaches

Recent research increasingly adopts multi-modal sensing, combining IMU and EMG signals to exploit complementary information. IMUs provide reliable kinematic event timing, while EMG reflects underlying muscle activation and motor intent.

Studies combining IMU and EMG have shown:

- improved gait phase classification accuracy
- better robustness to sensor noise
- enhanced interpretability of detected phases

Nevertheless, combining modalities introduces challenges related to data synchronization, differing sampling frequencies, and increased processing complexity. These challenges emphasize the importance of careful preprocessing and signal alignment, which are central aspects of this project.

3.6 Sit-to-Walk (STW) Transition Analysis

The STW transition has received increasing attention due to its relevance in fall-risk assessment and rehabilitation. Unlike steady-state walking, STW involves transient phases such as trunk flexion, seat-off, and gait initiation. Several studies segment STW into sub-phases using combinations of force plates, motion capture, IMUs, and EMG.

Common findings include:

- STW exhibits higher variability than steady walking
- event timing differs significantly across individuals
- gait initiation dynamics influence the first walking step

Many STW studies rely on laboratory-based equipment, limiting portability. Wearable-based STW analysis remains challenging, particularly for reliable phase segmentation without force plate data. This motivates approaches that first establish robust walking-phase detection before extending to STW-specific events.

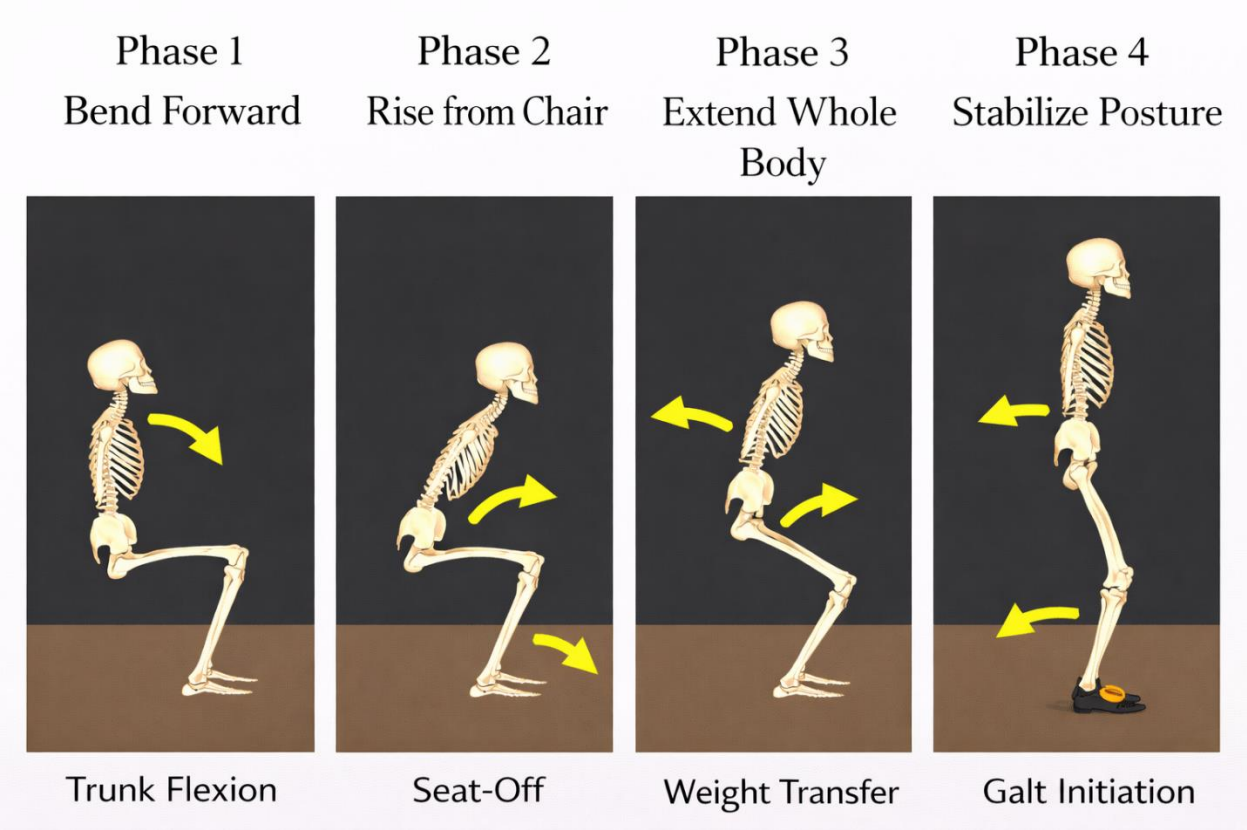


Figure 6: STW transition

3.7 Machine Learning for Gait Phase Classification

3.7.1 Classical Machine Learning Methods

Classical machine learning models such as Random Forests (RF), Support Vector Machine (SVM), k-Nearest Neighbors (kNN), and Linear Discriminant Analysis (LDA) are widely used for gait phase classification. These models typically operate on handcrafted features extracted from sensor signals.

Advantages of classical methods include:

- good performance on small to medium-sized datasets
- low computational requirements
- interpretability of feature importance (especially for RF)

Many studies report strong baseline performance using Random Forests and SVMs when features are carefully designed and normalized.

3.7.2 Deep Learning Approaches

More recent work explores deep learning models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Temporal Convolutional Networks (TCNs). These models can learn features directly from raw or minimally processed signals.

Despite their potential, deep learning approaches have limitations:

- require large labeled datasets
- increased training complexity
- reduced interpretability
- sensitivity to hyperparameter selection

For projects with limited data availability, feature-based classical methods often remain more practical and reliable.

3.8 Limitations in Existing Work

Based on the reviewed literature, several gaps and limitations can be identified:

- heavy reliance on laboratory-based ground truth systems
- limited discussion of data cleaning and non-finite value handling
- insufficient attention to multi-rate sensor synchronization
- inconsistent phase definitions across studies
- limited focus on STW transitions in wearable-only setups

These limitations highlight the need for transparent, end-to-end pipelines that explicitly address early-stage data handling and segmentation.

3.9 Positioning of This Project

The approach adopted in this project addresses several of the identified gaps by:

- focusing on a wearable-only solution using foot-mounted IMU and EMG
- explicitly handling multi-rate data and non-finite values
- using analytically defined gait events and reference points
- segmenting gait into phase-bounded windows for feature extraction
- building an ML-ready dataset with clear labeling logic

Rather than prioritizing model complexity, the project emphasizes robust preprocessing, segmentation, and feature extraction as prerequisites for reliable machine learning.

3.10 Summary

This chapter reviewed existing work on gait analysis, wearable sensor-based event detection, STW transitions, and machine learning approaches. While significant progress has been made, challenges remain in creating robust, portable, and reproducible gait phase detection systems using wearable sensors alone. The next chapter describes the dataset and experimental setup used in this project, laying the foundation for the proposed signal processing and segmentation pipeline.

Chapter 4 – Dataset and Experimental Setup

4.1 Chapter Overview

This chapter describes the dataset used in this project and the experimental setup under which the data were acquired. It details the sensor configuration, placement of EMG and IMU sensors, sampling characteristics, data file structure, and practical considerations related to data quality. A clear understanding of the dataset and acquisition setup is essential for interpreting the signal processing and segmentation methods presented in later chapters.

4.2 Dataset Description

The dataset used in this project consists of wearable sensor recordings collected during human locomotion tasks for 65 subject. The primary focus is on walking trials performed at different speeds, including 5 **slow**, 5 **normal**, and 5 **fast** walking conditions. In addition to that, there is a 3 MVC files for every subject, for the hamstring muscles, for the quads muscles, and for the shanks muscles. Each trial is stored as a CSV file containing synchronized recordings from multiple EMG channels and IMU sensors.

The recordings capture continuous time-series data rather than pre-segmented gait cycles. As a result, all gait events, phase boundaries, and labels must be derived algorithmically from the sensor signals. This dataset structure reflects realistic wearable-sensor scenarios, where raw signals are recorded without explicit ground-truth annotations.

Subject Descriptives			
Young Adults (19–35 Years)			
	Range	**Mean ± Standard Deviation**	
Age (Years)	19–33	24.19 ± 4.34	
Weight (kg)	38.60–107.90	64.12 ± 17.04	
Height (cm)	144.50–182.00	165.33 ± 9.45	
Middle-aged Adults (36–55 Years)			
Age (Years)	36–50	42.13 ± 4.55	
Weight (kg)	48.65–95.25	67.94 ± 15.85	
Height (cm)	144.80–179.40	164.46 ± 10.47	
Older Adults (56–73 Years)			
Age (Years)	57–73	64.00 ± 4.43	
Weight (kg)	45.20–84.10	59.38 ± 10.70	
Height (cm)	146.80–166.80	156.94 ± 6.07	
Subject Distribution by Biological Sex			
	Young Adults (19–35 Years)	**Middle-aged Adults (36–55 Years)**	**Older Adults (56–73 Years)**
Male	19	9	4
Female	13	6	14
Total	32	15	18

Table 1: Subject Descriptives

4.3 Sensor Configuration

4.3.1 Surface EMG Sensors

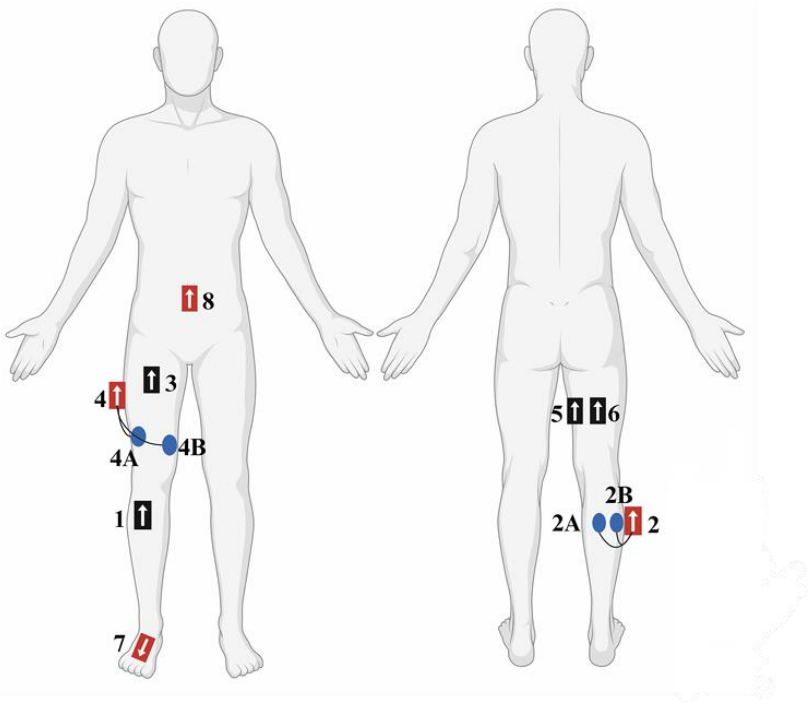
Surface electromyography (sEMG) signals are recorded from **eight lower-limb muscles** that are commonly involved in gait:

- Tibialis Anterior (TA)
- Gastrocnemius Medialis (GM)
- Gastrocnemius Lateralis (GL)

- Rectus Femoris (RF)
- Vastus Lateralis (VL)
- Vastus Medialis (VM)
- Semitendinosus (ST)
- Biceps Femoris (BF)

These muscles collectively represent ankle dorsi flexors and plantar flexors, knee extensors, and knee flexors, providing comprehensive coverage of lower-limb neuromuscular activity during walking.

Electrodes are assumed to be placed following standard anatomical guidelines to ensure signal consistency and minimize cross-talk. Although precise electrode placement details may vary, the dataset reflects typical EMG acquisition protocols used in gait analysis studies.



Legend	
	EMG & IMU
	IMU Only
	EMG Only
1	Tibialis Anterior
2	Gastrocnemius
2A	Gastrocnemius Medialis
2B	Gastrocnemius Lateralis
3	Rectus Femoris
4	Quadriceps
4A	Vastus Lateralis
4B	Vastus Medialis
5	Semitendinosus
6	Biceps Femoris
7	Foot IMU
8	Trunk IMU

Figure 7: sEMG locations

4.3.2 Inertial Measurement

In the provided dataset, various IMU sensors are employed, and they are placed at various body segments. These IMU sensors utilized are capable of collecting angular velocity and linear acceleration data. However, focusing on the **foot-mounted IMU** is crucial, especially when considering the reliable information for gait event detection.

Among all IMU sensors, the **gyroscope X-axis** of the foot IMU has been identified as the primary IMU signal for the detection of heel-strike, toe-off, and reference events such as mid-swing and mid-stance. Other IMUs, such as those located in the trunk, are not taken into consideration within the current analysis as they add complexity to the system.



Figure 8: The Sensors on a Random Subject Leg

4.4 Sampling Frequency and Multi-Rate Acquisition

A notable characteristic of the dataset is the presence of **multi-rate sampling** across different EMG channels. Specifically:

- TA, RF, ST, & BF are sampled at **1259 Hz**
- Other EMG channels are sampled at **1778 Hz**
- IMU data (including foot gyroscope signals) are sampled at **1259 Hz**

Multi-rate acquisition is common in practical systems due to hardware constraints and sensor-specific configurations. However, it introduces challenges for joint signal analysis, segmentation, and feature

extraction. Without resampling, signals from different channels cannot be aligned accurately in time, leading to inconsistent phase boundaries and unreliable feature computation.

To address this issue, all EMG channels sampled at 1778 Hz are resampled to match the reference sampling frequency of 1259 Hz. This choice ensures temporal alignment between EMG and IMU signals and allows the use of a single, consistent time vector throughout the processing pipeline.

4.5 Data File Structure

Each CSV file in the dataset contains a header followed by numerical data arranged in columns. The columns correspond to different sensor channels, including EMG signals and IMU signals from multiple sensor units.

Key characteristics of the file structure include:

- Fixed column positions for each sensor channel
- Separate column blocks for EMG and IMU data
- Presence of empty cells or non-numeric values in certain IMU columns at the beginning of recordings
- Trailing rows containing zeros, indicating the termination of valid data acquisition

For example, the foot IMU gyroscope Z-axis signal is located in a specific column associated with a designated sensor identifier and mode. The signal values begin after an initial sequence of empty entries, requiring identification of the first valid sample before further processing.

Understanding this structure is critical for correct data extraction and prevents misalignment or inclusion of invalid samples.

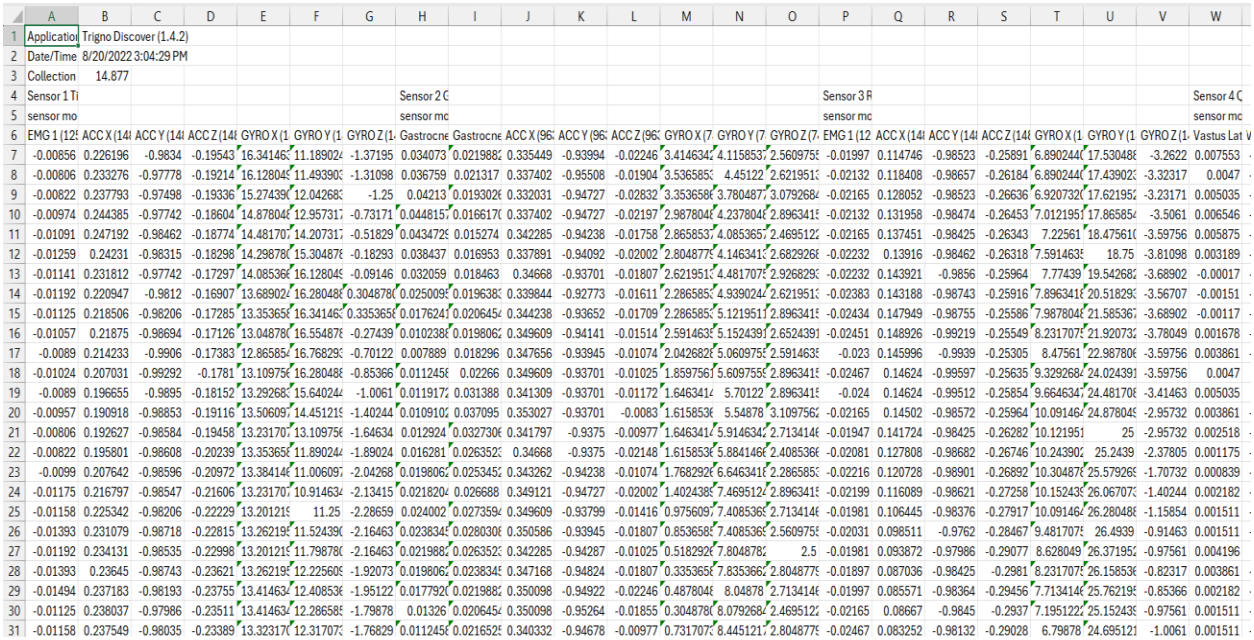
How the data saved in the PC:

On recording:

GaitEventsDetection

└── S08 (the subjects)
S09 …S75

└── walk_fast_1 (the trials)
walk_fast_2 …walk_slow_5



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	Applicatio	Trigno Discover (1.4.2)																					
2	Date/Time	8/20/2022 3:04:29 PM																					
3	Collection	14.877																					
4	Sensor 1 Ti							Sensor 2 C								Sensor 3 R							Sensor 4 C
5	sensor mo							sensor mc								sensor mc							sensor mc
6	EMG 1 (12	ACC X (14	ACC Y (14	ACC Z (14	GYRO X (1	GYRO Y (1	GYRO Z (1	Gastrocne	Gastrocne	ACC X (96	ACC Y (96	ACC Z (96	GYRO X (7	GYRO Y (7	GYRO Z (7	EMG 1 (12	ACC X (14	ACC Y (14	ACC Z (14	GYRO X (1	GYRO Y (1	GYRO Z (1	Vastus Lat
7	-0.00856	0.226196	-0.9834	-0.19543	16.34146	11.189024	-1.37195	0.034073	0.021988	0.335449	-0.93994	-0.02246	3.414634	4.115853	2.560975	-0.01997	0.114746	-0.98523	-0.25891	6.890244	17.53048	-3.2622	0.007553
8	-0.00806	0.233276	-0.97778	-0.19214	16.12804	11.493902	-1.31098	0.036759	0.021317	0.337402	-0.95508	-0.01904	3.536585	4.45122	2.621951	-0.02132	0.118408	-0.98657	-0.26184	6.890244	17.43902	-3.32317	0.0047
9	-0.00822	0.237793	-0.97498	-0.19336	15.27439	12.042683	-1.25	0.04213	0.019302	0.332031	-0.94727	-0.02832	3.353658	3.780487	3.079268	-0.02165	0.128052	-0.98523	-0.26636	6.920732	17.62195	-3.23171	0.005035
10	-0.00974	0.244385	-0.97742	-0.18604	14.87804	12.957317	-0.73171	0.044815	0.016617	0.337402	-0.94727	-0.02197	2.987804	4.237804	2.896341	-0.02132	0.131958	-0.98474	-0.26453	7.012195	17.86585	-3.5061	0.006546
11	-0.01091	0.247192	-0.98462	-0.18774	14.48170	14.207317	-0.51829	0.043472	0.015274	0.342285	-0.94238	-0.01758	2.865853	4.085365	2.469512	-0.02165	0.137451	-0.98425	-0.26343	7.22561	18.47561	-3.59756	0.005875
12	-0.01259	0.24231	-0.98315	-0.18298	14.29878	15.304878	-0.18293	0.038437	0.016953	0.337891	-0.94092	-0.02002	2.804877	4.146341	2.682926	-0.02232	0.13916	-0.98462	-0.26318	7.591463	18.75	-3.81098	0.003189
13	-0.01141	0.231812	-0.97742	-0.17297	14.08536	16.128049	-0.09146	0.032059	0.018463	0.34668	-0.93701	-0.01807	2.621951	4.481707	2.926829	-0.02232	0.143921	-0.9856	-0.25964	7.77439	19.54268	-3.68902	-0.00017
14	-0.01192	0.220947	-0.9812	-0.16907	13.68902	16.280488	0.304878	0.025095	0.019638	0.339844	-0.92773	-0.01611	2.286585	4.939024	2.621951	-0.02383	0.143188	-0.98743	-0.25916	7.896341	20.51829	-3.56707	-0.00151
15	-0.01125	0.218506	-0.98206	-0.17285	13.35365	16.341463	0.335365	0.017624	0.020645	0.344238	-0.93652	-0.01709	2.286585	5.121951	2.896341	-0.02434	0.147949	-0.98755	-0.25586	7.987804	21.58536	-3.68902	-0.00117
16	-0.01057	0.21875	-0.98694	-0.17126	13.04878	16.554878	-0.27439	0.010238	0.019806	0.349609	-0.94141	-0.01514	2.591463	5.152439	2.652439	-0.02451	0.148926	-0.99219	-0.25549	8.231707	21.92073	-3.78049	0.001678
17	-0.0089	0.214233	-0.9906	-0.17383	12.86585	16.768293	-0.70122	0.007889	0.018296	0.347656	-0.93945	-0.01074	2.042682	5.060975	2.591463	-0.023	0.145996	-0.9939	-0.25305	8.47561	22.98780	-3.59756	0.003861
18	-0.01024	0.207031	-0.99292	-0.1781	13.10975	16.280488	-0.85366	0.011245	0.02266	0.349609	-0.93701	-0.01025	1.859756	5.609755	2.896341	-0.02467	0.14624	-0.99597	-0.25635	9.329268	24.02439	-3.59756	0.0047
19	-0.0089	0.196655	-0.9895	-0.18152	13.29268	15.640244	-1.0061	0.011917	0.031388	0.341309	-0.93701	-0.01172	1.646341	5.70122	2.896341	-0.024	0.14624	-0.99512	-0.25854	9.664634	24.48170	-3.41463	0.005035
20	-0.00957	0.190918	-0.98853	-0.19116	13.50609	14.451219	-1.40244	0.010910	0.037095	0.353027	-0.93701	-0.0083	1.615853	5.54878	3.109756	-0.02165	0.14502	-0.98572	-0.25964	10.09146	24.87804	-2.95732	0.003861
21	-0.00806	0.192627	-0.98584	-0.19458	13.23170	13.109756	-1.64634	0.012924	0.032730	0.341797	-0.9375	-0.00977	1.646341	5.914634	2.713414	-0.01947	0.141724	-0.98425	-0.26282	10.12195	25	-2.95732	0.002518
22	-0.00822	0.195801	-0.98608	-0.20239	13.35365	11.890244	-1.89024	0.016281	0.026352	0.34668	-0.9375	-0.02148	1.615853	5.884146	2.408536	-0.02081	0.127808	-0.98682	-0.26746	10.24390	25.2439	-2.37805	0.001175
23	-0.0099	0.207642	-0.98596	-0.20972	13.38414	11.006097	-2.04268	0.019806	0.025345	0.343262	-0.94238	-0.01074	1.768292	6.646341	2.286585	-0.02216	0.120728	-0.98901	-0.26892	10.30487	25.57926	-1.70732	0.000839
24	-0.01175	0.216797	-0.98547	-0.21606	13.23170	10.914634	-2.13415	0.021820	0.026688	0.349121	-0.94727	-0.02002	1.402438	7.469512	2.896341	-0.02199	0.116089	-0.98621	-0.27258	10.15243	26.06707	-1.40244	0.002182
25	-0.01158	0.225342	-0.98206	-0.22229	13.20121	11.25	-2.28659	0.024002	0.027359	0.349609	-0.93799	-0.01416	0.975609	7.408536	2.713414	-0.01981	0.106445	-0.98376	-0.27917	10.09146	26.28048	-1.15854	0.001511
26	-0.01393	0.231079	-0.98718	-0.22815	13.26219	11.524390	-2.16463	0.023834	0.028030	0.350586	-0.93945	-0.01807	0.853658	7.408536	2.560975	-0.02031	0.098511	-0.9762	-0.28467	9.481707	26.4939	-0.91463	0.001511
27	-0.01192	0.234131	-0.98535	-0.22998	13.20121	11.798780	-2.16463	0.021988	0.026352	0.342285	-0.94287	-0.01025	0.518292	7.804878	2.5	-0.01981	0.093872	-0.97986	-0.29077	8.628049	26.37195	-0.97561	0.004196
28	-0.01393	0.23645	-0.98743	-0.23621	13.26219	12.225609	-1.92073	0.019806	0.023834	0.347168	-0.94824	-0.01807	0.335365	7.835366	2.804877	-0.01897	0.087036	-0.98425	-0.2981	8.231707	26.15853	-0.82317	0.003861
29	-0.01494	0.237183	-0.98193	-0.23755	13.41463	12.408536	-1.95122	0.017792	0.021988	0.350098	-0.94922	-0.02246	0.487804	8.04878	2.713414	-0.01997	0.085571	-0.98364	-0.29456	7.713414	25.76219	-0.85366	0.002182
30	-0.01125	0.238037	-0.97986	-0.23511	13.41463	12.286585	-1.79878	0.01326	0.020645	0.350098	-0.95264	-0.01855	0.304878	8.079268	2.469512	-0.02165	0.08667	-0.9845	-0.2937	7.195122	25.15243	-0.97561	0.001511
31	-0.01158	0.237549	-0.98035	-0.23389	13.32317	12.317073	-1.76829	0.011245	0.021652	0.340332	-0.94678	-0.00977	0.731707	8.445121	2.804877	-0.02467	0.083252	-0.98132	-0.29028	6.79878	24.69512	-1.0061	0.001511

Figure 9: Raw Data Excel File

On my Analysis:

GaitEventsDetection

└── S08

S09 … S75 (the subjects)

└── walk_fast_1 (the trials)

walk_fast_2 … walk_slow_5

├── 01_TO.xlsx (the events)

(include all TO readings from all the steps detected from that trial)

└── 02_MSw.xlsx

└── 03_HS.xlsx

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Step	Segment	Event	Time_s	TA	GM	GL	RF	VL	VM	ST	BF	TA_RMS	TA_IEMG	TA_MAV	TA_WL
2	1	1	MSw	0	0.038916885	0.065465976	0.04376429	0.035787772	0.009736835	0.163395508	0.008334716	0.006567859	0.093608808	18.2326876	0.068802595	7.5387908
3	1	1	MSw	0.00056243	0.058685248	0.084761406	0.051986357	0.067816821	0.018984382	0.07539354	0.007899963	0.001927749				
4	1	1	MSw	0.001124859	0.057705345	0.071346151	0.048175403	0.01724133	0.015598308	0.021244003	0.013204494	0.001828272				
5	1	1	MSw	0.001687289	0.04999815	0.039103857	0.029291125	0.07257028	0.019940109	0.113449787	0.017859831	0.001994553				
6	1	1	MSw	0.002249719	0.03880834	0.006868466	0.005042708	0.113998619	0.050345176	0.124985208	0.015992547	0.002439751				
7	1	1	MSw	0.002812148	0.021092278	0.018794612	0.00718856	0.111821217	0.046722605	0.058723734	0.00820822	0.007710689				
8	1	1	MSw	0.003374578	0.002777939	0.034552671	0.000357713	0.131378427	0.024773743	0.032930606	0.001036906	0.00741374				
9	1	1	MSw	0.003937008	0.024193532	0.03419763	0.014710759	0.158443109	0.01627383	0.099702131	0.000248115	0.000370907				
10	1	1	MSw	0.004499438	0.029651261	0.020104165	0.023068135	0.122381397	0.024780183	0.157333708	0.004951181	0.009352645				
11	1	1	MSw	0.005061867	0.015917885	0.007864341	0.021323549	0.037546385	0.030838675	0.124010937	0.008787484	0.010809333				
12	1	1	MSw	0.005624297	0.000858973	0.010817404	0.015699779	0.010013695	0.024586018	0.06887141	0.006845564	0.004134761				
13	1	1	MSw	0.006186727	0.004205859	0.025702598	0.010019899	0.001947887	0.014764932	0.084229529	0.001211081	0.001624402				
14	1	1	MSw	0.006749156	0.001348195	0.037933948	0.003466538	0.023324645	0.013791056	0.141092624	0.001185346	0.000212053				
15	1	1	MSw	0.007311586	0.010533145	0.037359214	0.001166598	0.030463956	0.023788605	0.154037504	0.002666774	0.005394814				
16	1	1	MSw	0.007874016	0.029771603	0.025459325	0.002867181	0.038182965	0.035039454	0.105568092	0.00726934	0.007219247				
17	1	1	MSw	0.008436445	0.041748673	0.009727674	0.01466728	0.045113879	0.037816542	0.039917069	0.006167818	0.006846503				
18	1	1	MSw	0.008998875	0.045396399	0.002830064	0.022466857	0.039475685	0.033762389	0.011737054	0.001280363	0.008070111				
19	1	1	MSw	0.009561305	0.053841596	0.00687996	0.01717218	0.031548456	0.027656258	0.054072414	0.001598506	0.008591183				
20	1	1	MSw	0.010123735	0.063113158	0.002217163	0.002533303	0.035123875	0.015326725	0.103026755	0.003487923	0.004622227				
21	1	1	MSw	0.010686164	0.055235671	0.002864343	0.011171616	0.035150853	0.004742518	0.154480136	0.008917121	0.001069112				
22	1	1	MSw	0.011248594	0.033073404	0.001031572	0.017284765	0.001126271	0.019831495	0.169994165	0.01456611	0.002950788				
23	1	1	MSw	0.011811024	0.018850331	0.002894739	0.015466172	0.061917886	0.021158207	0.116770541	0.013089388	0.000679843				
24	1	1	MSw	0.012373453	0.016195515	0.001361921	0.010547076	0.092960711	0.016904903	0.028267175	0.006706984	0.002269029				
25	1	1	MSw	0.012935883	0.0072511	0.001402433	0.01047342	0.047067943	0.012266392	0.02161345	0.004909456	0.004181881				
26	1	1	MSw	0.013498313	0.009032832	0.000512827	0.021266679	0.020591074	0.000136946	0.010707869	0.0099861	0.004847411				
27	1	1	MSw	0.014060742	0.009993662	0.001033148	0.040185464	0.014589872	0.016359293	0.016071457	0.01531233	0.003802678				
28	1	1	MSw	0.014623172	0.008971587	0.014258459	0.053911979	0.064854128	0.019686649	0.021687254	0.01562839	0.002369702				
29	1	1	MSw	0.015185602	0.015147681	0.026862963	0.049530857	0.135222836	0.006362457	0.020027393	0.010950911	0.003031611				
30	1	1	MSw	0.015748031	0.020587851	0.021938677	0.031528995	0.162461291	0.005846479	0.041420802	0.004242304	0.004649006				
31	1	1	MSw	0.016310461	0.084181912	0.006748944	0.019737827	0.175667983	0.005125179	0.077891143	0.000261255	0.003151042				
32	1	1	MSw	0.016872891	0.140607584	0.001900695	0.023302429	0.173059574	0.001139668	0.073821155	0.001086894	0.001954161				

Figure 10: Processed Data Excel File

4.6 Data Cleaning and Validity Checks

Real-world wearable datasets often contain imperfections that must be addressed prior to signal processing. In the present dataset, the following issues are observed:

1. **Non-finite values:**
   Certain channels may contain NaN or infinite values, particularly at the beginning of recordings or due to sensor initialization artifacts.

2. **Trailing zero segments:**
   After the end of valid acquisition, some channels contain rows where all values are zero. Including these segments in analysis would distort statistical features and event detection.

3. **Channel-specific termination:**
   Different sensor groups may terminate at slightly different times, requiring group-wise truncation.

To ensure numerical stability and signal validity:

- Non-finite values are replaced or removed prior to filtering.
- Each group of channels is truncated at the first occurrence of all-zero rows.
- Only the continuous, valid portion of the signal is retained for further processing.

These steps are essential to prevent filtering failures and to ensure that extracted features reflect actual physiological activity.

4.7 Experimental Protocol

The dataset includes walking trials performed under different speed conditions. Although the exact experimental protocol may vary, the general procedure consists of:

1. Subject preparation and sensor placement
2. Recording of continuous EMG and IMU signals
3. Execution of walking trials at predefined speeds
4. Storage of raw sensor data without manual segmentation

No external ground-truth labeling (e.g., force plates or motion capture) is provided. As a result, all gait events and phase boundaries are derived from wearable sensor signals alone. This setup reflects practical constraints encountered in wearable gait analysis and motivates the algorithmic approach adopted in this project.

4.8 Assumptions and Constraints

Several assumptions are made in order to process and analyze the dataset:

- Sensor placement is assumed to be consistent across trials.
- Walking trials are assumed to be continuous and free of abrupt stops.
- Variability across walking speeds is expected and treated as part of the dataset.

These assumptions are consistent with common practice in wearable gait analysis and are explicitly acknowledged to maintain transparency.

4.9 Recap

This chapter introduced the dataset and experimental setup used in the project, such as sensor configuration, characteristics of sampling, data file structure, and considerations on data quality. Particular attention was drawn to challenges related to multi-rate acquisition and imperfect real-world recordings. The following chapter outlines the signal processing pipeline for the presented dataset by providing a detailed overview of the methods adopted to perform synchronization, filtering, and to prepare the EMG and IMU signals for gait event detection and segmentation.

Chapter 5 – Signal Processing Pipeline

5.1 Chapter Overview

In this chapter, the complete signal processing pipeline used prior to gait event detection, phase segmentation, and feature extraction on raw sensor signals will be illustrated. The purpose of signal processing in this chapter shall be to take unclear and varied rate raw signals of EMG and IMU signals and turn them into high-quality clean signals that are synchronized and ready to use.

The pipeline is designed to be modular and reproducible, allowing individual steps to be validated independently and facilitating later extension to other locomotion tasks such as Sit-to-Walk (STW).

5.2 Raw Signal Inspection and Preprocessing Rationale

Before applying any filtering or transformations, the raw signals are visually inspected to assess signal quality and identify artifacts. EMG signals typically exhibit high-frequency stochastic behavior with occasional spikes due to motion artifacts, while IMU gyroscope signals show smoother, lower-frequency patterns associated with limb motion.

Initial inspection reveals several common issues:

- Presence of non-finite values (NaN or Inf)
- Empty samples at the beginning of certain IMU channels
- Trailing zero segments indicating the end of valid recording
- Differences in sampling frequency across EMG channels

These observations motivate the need for a carefully ordered preprocessing sequence, as applying filters or feature extraction directly to raw data would lead to numerical instability and unreliable results.

5.3 Handling Non-Finite and Invalid Samples

5.3.1 Non-Finite Values

Filtering operations such as zero-phase digital filtering require input signals to contain only finite values. Non-finite samples cause numerical errors and may lead to complete processing failure. Therefore, all signals are checked for NaN and infinite values prior to filtering.

Non-finite samples are handled by:

- replacing isolated non-finite values with zeros or interpolated values
- removing initial empty segments until the first valid sample is encountered

This step ensures numerical stability and prevents filter divergence.

5.3.2 Trailing Zero Truncation

After the completion of data acquisition, several channels contain rows of all-zero values. These segments do not represent physiological activity and would distort statistical feature calculations and event detection if included.

For each group of signals, the data are truncated at the first row where all values are zero, retaining only the continuous segment corresponding to valid acquisition.

5.4 Signal Synchronization and Resampling

5.4.1 Multi-Rate EMG Signals

As described in Chapter 4, the dataset includes EMG channels sampled at two different frequencies: 1259 Hz and 1778 Hz. Since gait event detection and phase segmentation rely on consistent temporal alignment, all channels must share a common time base.

5.4.2 Resampling Strategy

In this project, 1259 Hz is selected as the reference sampling frequency, as it matches the IMU sampling rate. EMG channels sampled at 1778 Hz are resampled to 1259 Hz using polyphase resampling techniques.

This approach preserves signal integrity while ensuring that:

- all EMG channels are time-aligned
- EMG and IMU signals share a common time vector
- phase boundaries can be applied consistently across all channels

5.5 EMG Signal Processing

5.5.1 Band-Pass Filtering

Raw EMG signals are contaminated by low-frequency motion artifacts and high-frequency noise. To isolate the physiologically relevant frequency band, each EMG channel is band-pass filtered.

A typical EMG band-pass range of **20–450 Hz** is applied, which:

- attenuates motion artifacts below 20 Hz, which is the STW stage
- suppresses high-frequency noise above 450 Hz
- preserves muscle activation information

Zero-phase filtering is used to prevent phase distortion and maintain accurate temporal alignment with gait events.

5.5.2 Notch Filtering

Powerline interference is a common artifact in EMG recordings. To suppress this interference, a notch filter centered at **50 Hz** is applied to each EMG channel. This step reduces spectral contamination without significantly affecting the underlying EMG content.

5.5.3 Rectification and Envelope Preparation

Following filtering, EMG signals are full-wave rectified by taking the absolute value of the signal. Rectification converts the bipolar EMG waveform into a unipolar representation that reflects muscle activation intensity.

Depending on the analysis objective, rectified EMG may be further smoothed using low-pass filtering or moving window techniques to obtain an envelope. In this project, rectified EMG signals are primarily used for feature extraction rather than event detection, and envelope smoothing is applied selectively when required for visualization.

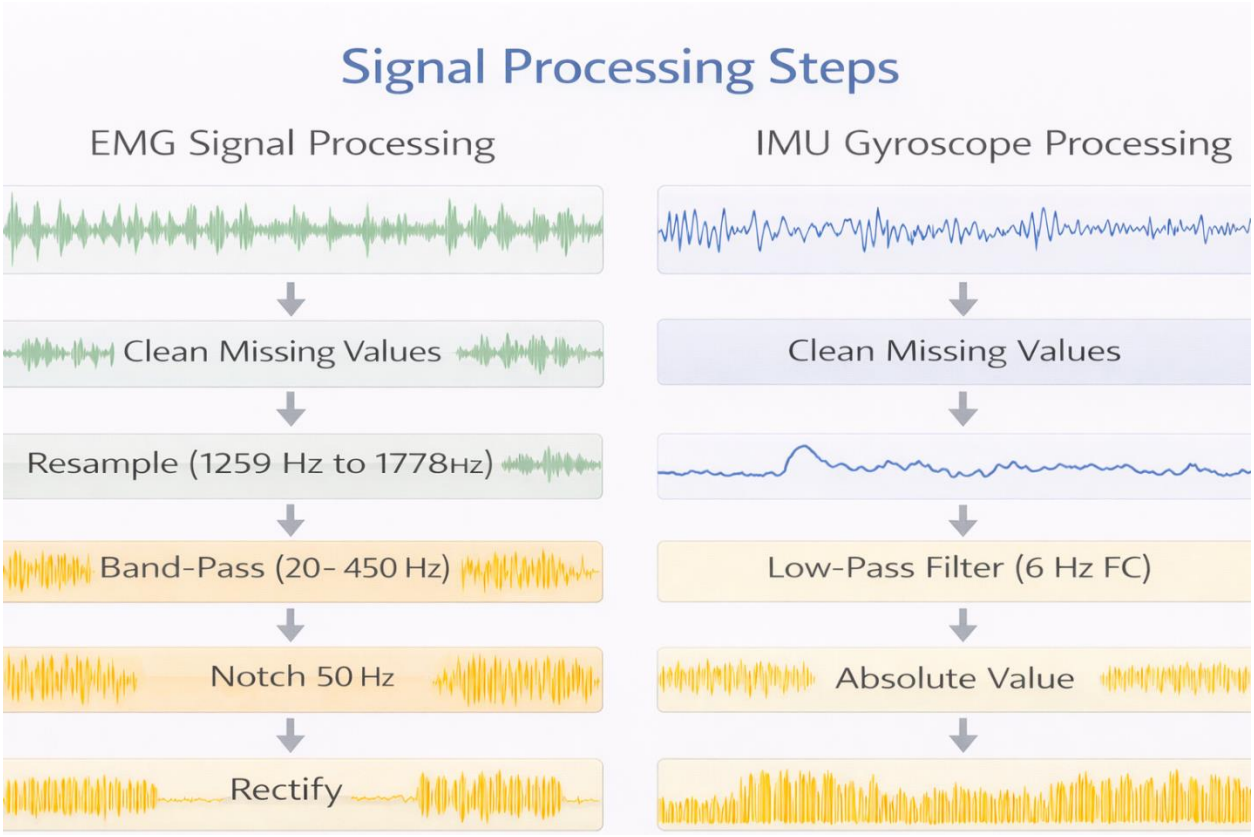


Figure 11: Signal Processing Steps

5.6 IMU Gyroscope Signal Processing

5.6.1 Selection of Gyroscope Axis

The gyroscope X-axis of the foot-mounted IMU is the axis captures the dominant rotational motion of the foot during walking and exhibits characteristic patterns during heel-strike, toe-off, and swing.
The IMU position on the foot as the Z-axis on the figure, so the rotational is happening on the X-axis.

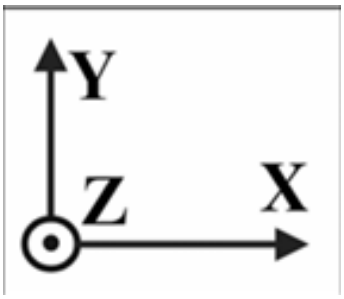


Figure 12: Foot IMU Axis Orientation

5.6.2 Filtering of Gyroscope Signals

Raw gyroscope signals contain high-frequency noise that may interfere with peak detection. To emphasize gait-related motion dynamics, the gyroscope X-axis signal is filtered using:

- a low-pass filter with a cutoff around 6 Hz,

Zero-phase filtering is applied to maintain accurate timing of detected events.

5.7 Signal Normalization

To reduce inter-channel and inter-trial variability, signals may be normalized prior to feature extraction. Normalization ensures that features extracted from different channels are comparable and prevents features with large absolute magnitudes from dominating machine learning models.

In this project, normalization is performed using Maximum Voluntary Contraction (MVC) normalization, when reference values are available. MVC normalization scales EMG signals relative to a muscle-specific maximum activation level. Let $x(t)$denote the preprocessed EMG signal of a given muscle. The normalized signal is defined as:

$$x_{\text{norm}}(t) = \frac{x(t)}{\text{MVC}} \quad \ldots (7)$$

where MVC represents the maximum EMG amplitude recorded for the corresponding muscle during a maximal voluntary contraction trial, (which is came from taking the maximum value for each muscle from the 3 MVC files then take the maximum of them). This process completed for every subject.

Normalization is applied at the feature level during machine learning dataset construction in order to preserve raw signal characteristics during gait event detection and phase segmentation.

5.8 Verification Through Visualization

At each stage of the signal processing pipeline, visual inspection is used to verify correctness. Representative plots are generated to compare:

- raw versus filtered EMG signals
- rectified EMG waveforms
- raw versus filtered gyroscope signals

Visualization serves as an essential diagnostic tool to confirm that filtering parameters are appropriate and that no physiologically meaningful information is lost during preprocessing.

5.9 Pipeline Summary

The signal processing pipeline described in this chapter converts raw, imperfect wearable sensor recordings into synchronized and conditioned signals suitable for gait event detection and segmentation. The key steps include data cleaning, handling of non-finite values, resampling of multi-rate EMG channels, filtering of EMG and IMU signals, and MVC normalization.

By establishing a robust preprocessing foundation, this pipeline ensures that subsequent gait event detection, phase segmentation, and feature extraction are based on reliable and physiologically meaningful signals.

The next chapter describes the methods used to detect gait events, segment the gait cycle into phases, and extract features for machine learning dataset construction.

Chapter 6 – Segmentation and Feature Extraction

6.1 Chapter Overview

This chapter describes the methodology used to segment continuous walking data into physiologically meaningful gait phases and to extract features from each segmented phase for machine learning dataset construction. Building upon the signal processing pipeline presented in Chapter 5 and the theoretical foundations introduced in Chapter 2, this chapter translates biomechanical definitions into a practical and robust implementation.

The primary objectives of this chapter are:

- to detect gait events from foot-mounted IMU signals,

- to segment each gait cycle into distinct phases,
- to extract time- and frequency-domain features from EMG and IMU signals within each phase window,
- and to assemble a labeled dataset suitable for supervised machine learning.

6.2 Gait Event Detection Strategy

6.2.1 Selected Signal for Event Detection

Gait event detection is performed using the **gyroscope X-axis signal** of the foot-mounted IMU. This signal captures the dominant rotational motion of the foot during walking and exhibits consistent extrema associated with heel-strike, toe-off, and swing dynamics.

The choice of a single foot-mounted IMU simplifies the system architecture while maintaining sufficient robustness for gait event detection, as supported by the literature reviewed in Chapter 3.

while the resampling frequences happening, all the signals became at the same frequency so at this level when I found the exact time when the events happening for the IMU-foot, I can detect the events for the EMG signals based on that exact time.

6.2.2 Reference Events for Temporal Anchoring

To improve robustness and reduce ambiguity in noisy signals, reference events are identified prior to detecting fundamental gait events. These reference events include:

- **Mid-Swing (MSw):** the point of maximum angular velocity during swing
- **Mid-Stance (MSt):** the quasi-static point during stance where angular velocity approaches zero

These reference events serve as temporal anchors that constrain the search windows for heel-strike and toe-off detection.

6.2.3 Detection of Heel-Strike and Toe-Off

Heel-strike (HS) and toe-off (TO) are detected based on the analytical definitions introduced in Chapter 2. In practice, the detection process follows a constrained search approach:

1. Identify mid-swing events as prominent maxima in the gyroscope signal.
2. Search for heel-strike as a local minimum occurring after mid-swing and before the next stance phase.
3. Search for toe-off as a local minimum occurring between the previous heel-strike and mid-swing, with additional amplitude constraints to differentiate it from heel-strike.

This constrained strategy reduces false detections caused by spurious extrema and ensures temporal consistency across gait cycles.

Due to noise, sensor imperfections, and inter-subject variability inherent in wearable sensor data, the analytical equations defining gait events are not applied as strict equality conditions. Instead, they are implemented using adaptive thresholds and temporally constrained search windows relative to reference events such as mid-swing and mid-stance. This strategy preserves the biomechanical interpretation of gait events while ensuring robust and repeatable detection across different walking speeds and trials.

6.3 Practical Implementation Considerations

Although gait event definitions were expressed analytically in Chapter 2, direct application of these equations to real-world IMU signals is not feasible without additional constraints. Wearable sensor data are affected by noise, sensor bias, inter-subject variability, and changes in walking speed.

Therefore, the theoretical formulations are implemented using:

- adaptive thresholds rather than strict equality conditions,
- temporal search windows defined relative to reference events,
- tolerance margins to account for near-zero angular velocity conditions.

This approach preserves biomechanical interpretation while ensuring robust and repeatable event detection across different trials and walking speeds.

6.4 Gait Cycle Definition

A **gait cycle** is defined as the interval between two consecutive heel-strike events of the same foot. Once heel-strike events are detected, the duration of each gait cycle is computed as:

$$T_{\text{stride}} = t_{HS}^{(i+1)} - t_{HS}^{(i)} \quad \ldots (8)$$

Only complete gait cycles are retained for further analysis. Partial cycles occurring at the beginning or end of a recording are excluded to ensure consistency in segmentation and feature extraction.

6.5 Gait Phase Segmentation

6.5.1 Phase Definition

Each detected gait cycle is segmented into **four gait phases**:

- **Loading Response (LR):** from heel-strike to foot flat
- **Mid-Stance (MS):** from foot flat to heel off

- **Terminal Stance (TS):** from heel off to toe-off
- **Swing (SW):** from toe-off to the next heel-strike

These phase boundaries are derived from detected gait events (HS, FF, HO, TO). In cases where secondary events (FF, HO) are not reliably detectable, phase boundaries are approximated using stride-relative timing constraints derived from biomechanical literature.

6.5.2 Handling Event Uncertainty

Secondary gait events such as foot flat and heel off are inherently more difficult to detect than heel-strike and toe-off. To handle this uncertainty:

- temporal constraints are applied to ensure correct event ordering,
- implausible phase durations are rejected,
- fallback timing-based approximations are used when required.

This strategy ensures consistent phase segmentation without introducing unstable or physiologically implausible boundaries.

6.6 Phase-Bounded Window Extraction

Once phase boundaries are established, the continuous EMG and IMU signals are segmented into **phase-bounded windows**. Each window corresponds to a single gait phase within a specific stride.

Phase-bounded segmentation provides several advantages over fixed-length sliding windows:

- alignment with biomechanical events,
- improved interpretability of extracted features,
- reduced mixing of different movement dynamics within a single feature vector.

Each phase window is treated as an independent sample for feature extraction.

6.7 Feature Extraction Methodology

Feature extraction is performed on phase-bounded signal segments obtained from the gait segmentation process described earlier in this chapter. For each gait phase, features are extracted independently from each EMG channel. The extracted features are designed to capture signal amplitude, complexity, oscillatory behavior, and spectral characteristics relevant to gait phase discrimination.

Let $x(n)$ denote a discrete signal segment of length N, corresponding to a specific gait phase.

6.7.1 Time-Domain Features

Time-domain features are computationally efficient and widely used in EMG- and IMU-based gait analysis due to their interpretability and robustness. The following features are extracted from each phase window.

1. **Root Mean Square (RMS)**
   RMS reflects the effective signal power and muscle activation intensity:

$$RMS = \sqrt{\frac{1}{N}\sum_{n=1}^{N} x^2(n)} \quad \dots (9)$$

2. **Mean Absolute Value (MAV)**
   MAV represents the average amplitude of the signal:

$$MAV = \frac{1}{N}\sum_{n=1}^{N} | x(n) | \quad \dots (10)$$

3. **Integrated EMG (IEMG)**
   IEMG quantifies the total muscle activation within a phase:

$$IEMG = \sum_{n=1}^{N} | x(n) | \quad \dots (11)$$

4. **Waveform Length (WL)**
   WL captures the cumulative variation of the signal and reflects both amplitude and frequency characteristics:

$$WL = \sum_{n=1}^{N-1} | x(n+1) - x(n) | \quad \dots (12)$$

5. **Zero Crossings (ZC)**
   ZC estimates frequency-related behavior by counting sign changes that exceed a noise threshold ε:

$$\mathrm{ZC} = \sum_{n=1}^{N-1} f(x(n), x(n+1)) \quad \dots (13)$$

where:

$$f = \begin{cases} 1, & \text{if } x(n) \cdot x(n+1) < 0 \ \cap \ |\, x(n) - x(n+1) \,| \geq \epsilon \\ 0, & \text{otherwise} \end{cases}$$

6. **Slope Sign Changes (SSC)**
   SSC measures signal oscillation by detecting changes in slope direction:

$$\text{SSC} = \sum_{n=2}^{N-1} g\big(x(n-1), x(n), x(n+1)\big) \quad \ldots (14)$$

where:

$$g = \begin{cases} 1, & \text{if } (x(n) - x(n-1))(x(n) - x(n+1)) \geq \epsilon \\ 0, & \text{otherwise} \end{cases}$$

where $1\{\cdot\}$ denotes the indicator function.

6.7.2 Frequency-Domain Features

In addition to time-domain features, frequency-domain features are extracted to characterize the spectral properties of the signal within each gait phase. Power spectral density (PSD) estimation is used to obtain frequency-domain representations.

Let $P(f)$ denote the PSD of the signal.

1. **Peak Frequency (PF)**
   PF corresponds to the dominant frequency component:

$$PF = arg \max_{j=1,\ldots,M} (P_{xx}(f_j)) \quad \ldots (15)$$

   where:
   - $P_{xx}(f_j)$ is the power spectral density estimated using Welch's method,
   - f_j is the corresponding frequency bin,
   - M is the number of frequency bins.

2. **Total Power (TP)**
   TP represents the total spectral energy within a defined frequency band:

$$TP = \int_{f_1}^{f_M} P_{xx}(f)\, df \quad \ldots(16)$$

- This is the **mathematical definition** of Total Power.
- Used in theory, textbooks, and signal-processing derivations.
- Assumes a **continuous frequency spectrum**.

This is what Total Power means.

$$TP \approx \sum_{j=1}^{M-1} \frac{P_{xx}(f_j) + P_{xx}(f_{j+1})}{2} \left(f_{j+1} - f_j\right) \quad \ldots(17)$$

- This is a **numerical approximation** of the integral.
- This is **exactly what MATLAB's trapz(f,Pxx) computes**.
- Used because real data are **discrete**, not continuous.

This is how Total Power is actually computed in your code.

6.7.3 Feature Scope and Application

All features are computed separately for each gait phase (LR, MS, TS, SW), ensuring that the resulting feature vectors are phase-specific and biomechanically meaningful. This phase-bounded approach avoids mixing heterogeneous movement dynamics and improves class separability in subsequent machine learning models.

6.8 Feature Vector Construction

For each phase window, features from all EMG channels are concatenated to form a single feature vector. Let:

- N_m denote the number of EMG channels,
- N_f denote the number of features per channel.

The dimensionality of each feature vector is:

$$D = N_m \times N_f \quad \ldots(18)$$

6.9 Label Assignment and Dataset Structure

Each feature vector is assigned a class label corresponding to its gait phase:

$$y \in \{LR, MS, TS, SW\}$$

The final dataset is structured as:

- X: feature matrix $\in \mathrm{R}^{N \times D}$
- y: phase label vector $\in \mathrm{R}^{N}$

Additional metadata such as stride index and walking speed may be included for analysis and evaluation purposes.

6.10 Quality Control and Sample Validation

To ensure dataset integrity, several quality control checks are applied:

- verification of correct event ordering within each stride,
- rejection of samples with abnormal phase durations,
- inspection of feature distributions for outliers.

These steps reduce noise in the training data and improve machine learning performance.

6.11 Summary

This chapter presented the methods used to detect gait events, segment gait cycles into physiologically meaningful phases, and extract features for machine learning dataset construction. By combining biomechanical definitions with practical implementation strategies, the proposed approach achieves robust segmentation suitable for wearable-sensor-based gait analysis.

The resulting labeled dataset forms the basis for machine learning model training and evaluation, which are described in the next chapter.

Chapter 7 – ML Models & Evaluation

7.1 Introduction

This chapter presents the machine learning approaches used for automated gait event classification and prediction during the Sit-to-Walk (STW) transition. Based on the extracted EMG features and segmented gait phases (TO, MSw, HS), two complementary machine learning models were implemented:

- **Random Forest (RF)** – for event classification using engineered features.

- **Long Short-Term Memory (LSTM)** network – for sequence-based next-event prediction.

The combination of these two models allows both static feature-based classification and dynamic temporal modeling of gait progression.

7.2 Feature-Based Classification Using Random Forest

7.2.1 Motivation for Using Random Forest

Random Forest was selected as the first machine learning model for several reasons:

1. **Robustness to Noise**
   EMG signals are inherently noisy and subject-dependent. Random Forest handles noisy and non-linear data effectively.
2. **Non-Parametric Model**
   RF does not assume normality or linear separability of features.
3. **Handles High-Dimensional Features Well**
   Each segment contains 64 features (8 muscles × 8 features), and RF performs well in medium-to-high dimensional spaces.
4. **Feature Importance Estimation**
   RF provides predictor importance scores, which are useful for:
   - Identifying dominant muscles
   - Identifying dominant EMG features (RMS, WL, ZC, etc.)
   - Physiological interpretation
5. **Fast Training and Stable Performance**
   Compared to deep learning models, RF requires less tuning and computational resources.

7.2.2 Input and Output Structure (RF)

Each gait segment (TO, MSw, HS) is represented by:

- **Input vector (64 features):**

  RMS, IEMG, MAV, WL, ZC, SSC, Peak Frequency (PF), Total Power (TP)
  for each of the 8 muscles.

- **Output label:**
  - 1 → TO
  - 2 → MSw
  - 3 → HS

Thus, the RF performs **multi-class classification** of the current gait event.

7.2.3 Training and Evaluation Strategy

A subject-wise split was applied to ensure generalization:

- First 55 subjects → Training set
- Remaining 10 subjects → Test set

This prevents data leakage between subjects and ensures evaluation on unseen individuals.

Model parameters:

- Method: Bagging
- Base learner: Decision Trees
- Number of trees: 50
- Evaluation: Confusion matrix and accuracy

The RF achieved high classification accuracy (~90–91.91%), demonstrating strong separability between gait events using engineered EMG features.

7.2.4 Strengths and Limitations of RF

Strengths:

- Interpretable
- Computationally efficient
- Strong baseline performance
- Provides feature importance

Limitations:

- Does not model temporal dependency
- Treats each segment independently
- Cannot inherently predict next event transitions

This limitation motivated the use of a sequence-based model.

7.3 Sequence Modeling Using LSTM

7.3.1 Motivation for Using LSTM

While Random Forest classifies individual segments, gait is inherently sequential:

TO → MSw → HS → TO → ...

To model this temporal dependency, a **Long Short-Term Memory (LSTM)** network was implemented.

LSTM was chosen because:

1. **Captures Temporal Dependencies**
   Gait events follow structured transitions.
2. **Handles Variable-Length Sequences**
   Each trial contains different numbers of steps.
3. **Remembers Long-Term Context**
   LSTM maintains memory of previous gait events.
4. **Suitable for Biomedical Time-Series**
   LSTM is widely used in EMG and motion analysis research.

7.3.2 Input and Output Structure (LSTM)

Unlike RF, the LSTM operates on sequences.

For each trial:

Input:

A sequence of feature vectors: [64 features × time steps]

Each time step corresponds to one detected gait segment.

Output:

Next event prediction:

Given current event → Predict next event

Training pairs are constructed using sequence shifting:

Input:

TO, MSw, HS, TO, ...

Target:

MSw, HS, TO, ...

Thus, the LSTM performs **sequence-to-sequence classification**.

7.3.3 LSTM Architecture

The implemented network consists of:

- Sequence Input Layer (64 features)
- LSTM Layer (128 hidden units, Output Mode = 'sequence')
- Fully Connected Layer (3 classes)
- Soft max Layer
- Classification Layer

Training configuration:

- Optimizer: Adam
- Epochs: 40
- Mini-batch size: 32
- Subject-wise split (55 train / 10 test)

7.3.4 Evaluation Strategy

Evaluation was performed on unseen subjects only.

Metrics:

- Overall accuracy
- Confusion matrix
- Class-wise performance

The LSTM demonstrates improved modeling of event transitions compared to RF, especially in predicting the next gait phase.

7.4 Final Model Strategy

The models serve complementary roles:

- RF → Strong baseline classifier for current event detection.
- LSTM → Advanced sequential predictor for next event forecasting.

This dual-model approach allows both robust classification and dynamic temporal prediction during the Sit-to-Walk transition.

7.5 Conclusion

This chapter demonstrated that:

- Engineered EMG features are sufficient for accurate gait event classification using Random Forest.
- Sequential modeling via LSTM improves temporal prediction and better reflects physiological gait progression.
- Subject-wise splitting ensures reliable generalization.

The combined approach provides a reliable framework for intelligent gait event detection and prediction.

Chapter 8 – Results, Discussion, Limitations

8.1 Results

8.1.1 Random Forest Classification Results

Overall Performance

The Random Forest (RF) model was trained using 50 decision trees under a subject-wise split (55 training subjects, 10 testing subjects). The resulting classification performance on unseen subjects demonstrated strong discriminative capability.

Overall Accuracy

= **91.91%**

Class-Wise Metrics:

Class	Precision	Recall	F1-score
TO	0.960	0.958	0.959
MSw	0.901	0.894	0.897
HS	0.898	0.905	0.902

Table 2: RF Class-Wise Metrics

Most misclassifications occurred between adjacent gait phases, which is physiologically expected because:

- Muscle activation patterns overlap near transition boundaries.
- MSw and HS share partial activation characterist

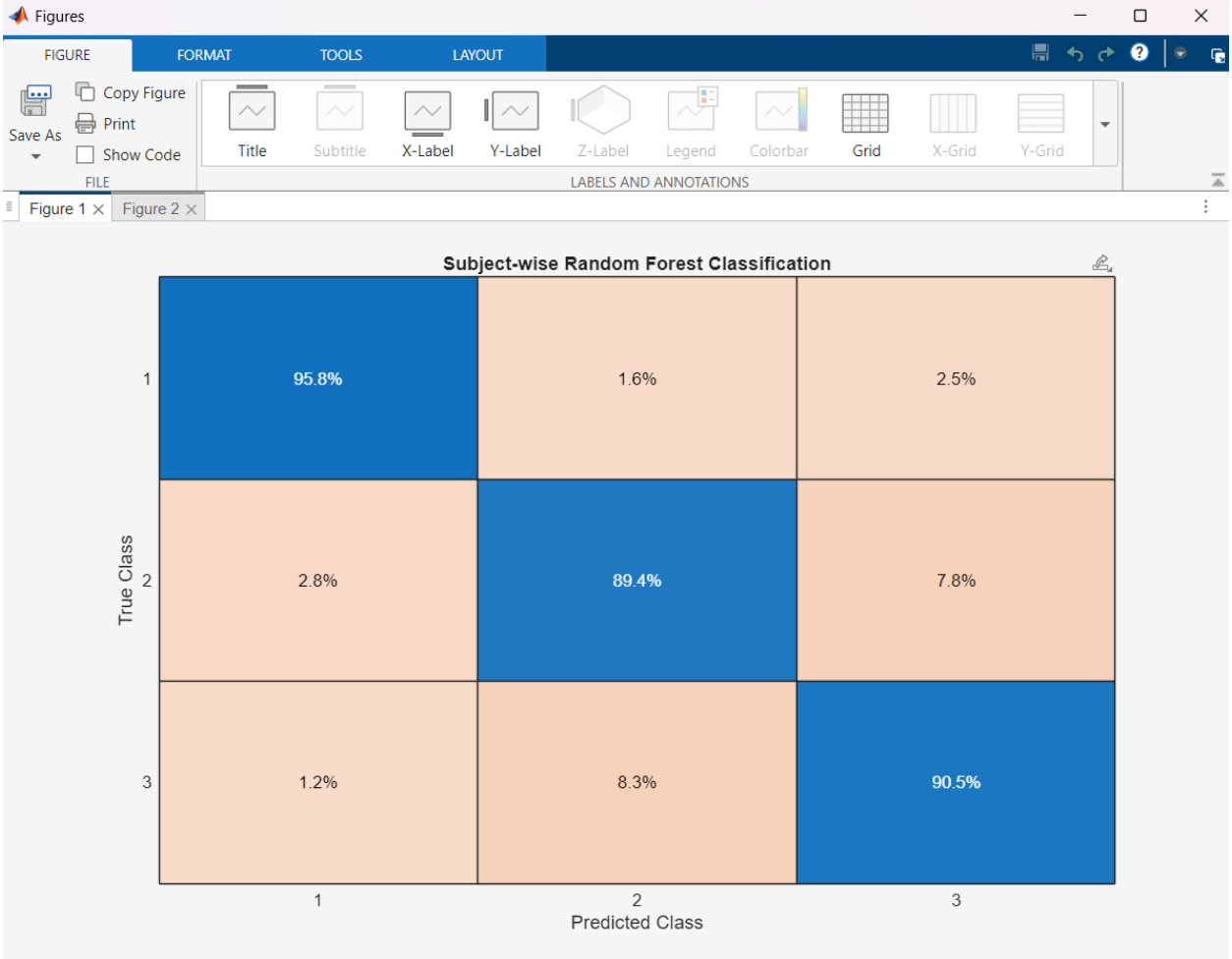


Figure 13: RF Confusion Matrix

Class-by-Class Analysis:

Class 1 (True Class 1)

- **95.8% correctly classified**
- 1.6% misclassified as Class 2
- 2.5% misclassified as Class 3

Class 2 (True Class 2)

- **89.4% correctly classified**
- 2.8% predicted as Class 1
- 7.8% predicted as Class 3

Class 3 (True Class 3)

- **90.5% correctly classified**
- 1.2% predicted as Class 1
- 8.3% predicted as Class 2

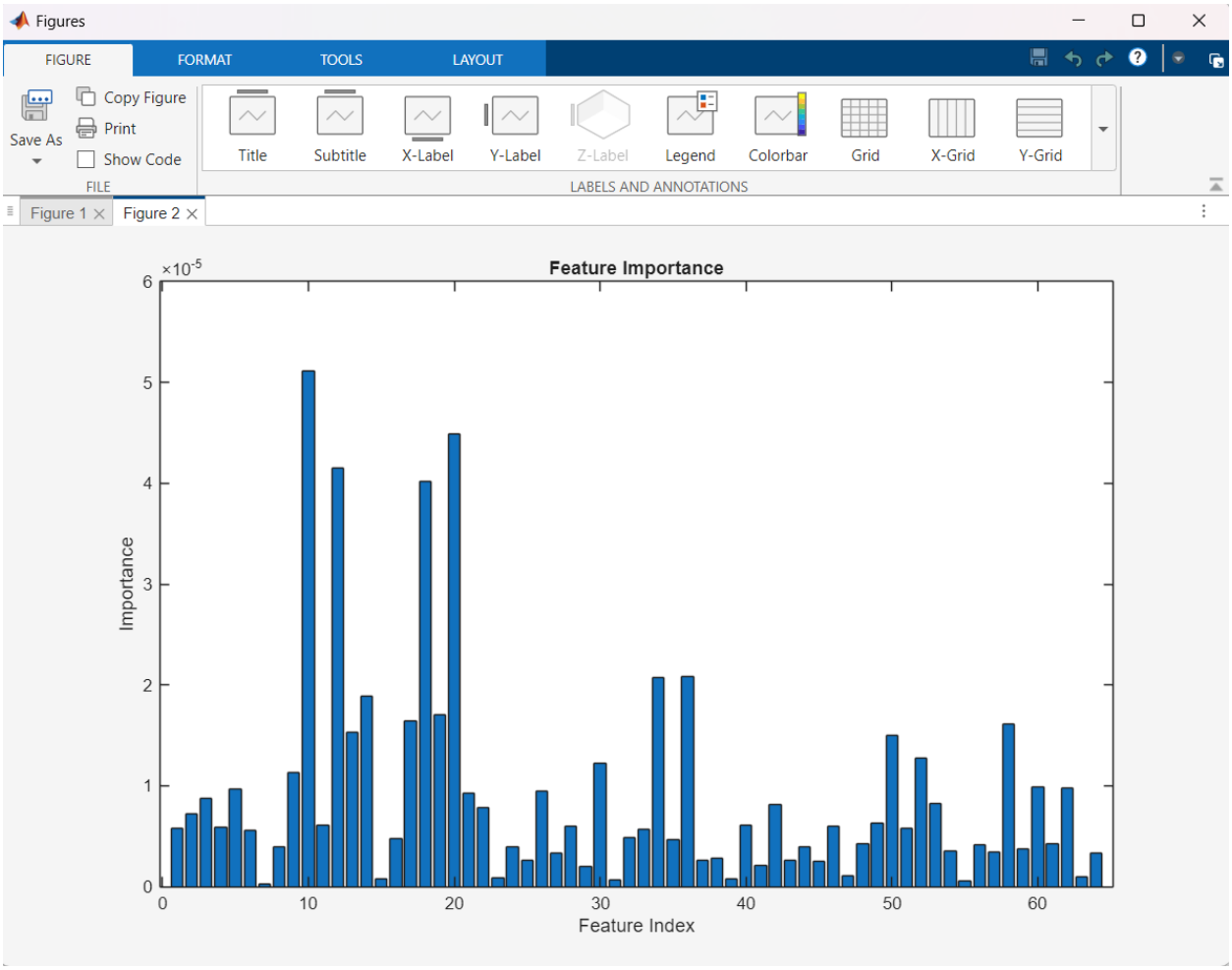


Figure 14: RF Feature Importance Plot

8.1.2 Random Forest Feature Importance

The feature importance analysis reveals that:

- Certain time-domain features (especially RMS and WL across specific muscles) contributed significantly.
- A subset of features (not uniformly distributed across all 64 inputs) dominated predictive performance.
- Some frequency-domain features (PF and TP) showed moderate influence but were not universally dominant.

This suggests:

- Time-domain EMG amplitude and variability measures are more discriminative than pure spectral measures for event detection.

- Not all muscles contribute equally to phase classification.

The feature importance distribution supports physiological expectations that certain muscle groups are more active during specific gait events.

8.1.3 LSTM Next-Event Prediction Results

Unlike the RF model, which performs static classification, the LSTM model was designed to predict the **next gait event** using sequential modeling.

Training Behavior

From the training curves:

- Accuracy rapidly increased from ~30% to >95% within early epochs.
- Final training accuracy stabilized around **~97–98%**.
- Training loss decreased smoothly and converged below 0.1.
- No instability or oscillatory divergence was observed.

This indicates:

- Stable optimization using Adam.
- Effective learning of temporal dependencies.
- No immediate signs of training instability.

Overall next-event prediction accuracy:

= **92.03 %**

Class-Wise Metrics:

Class	Precision	Recall	F1-score
TO	0.906	0.969	0.936
MSw	0.923	0.896	0.909
HS	0.932	0.902	0.917

Table 3: LSTM Class-Wise Metrics

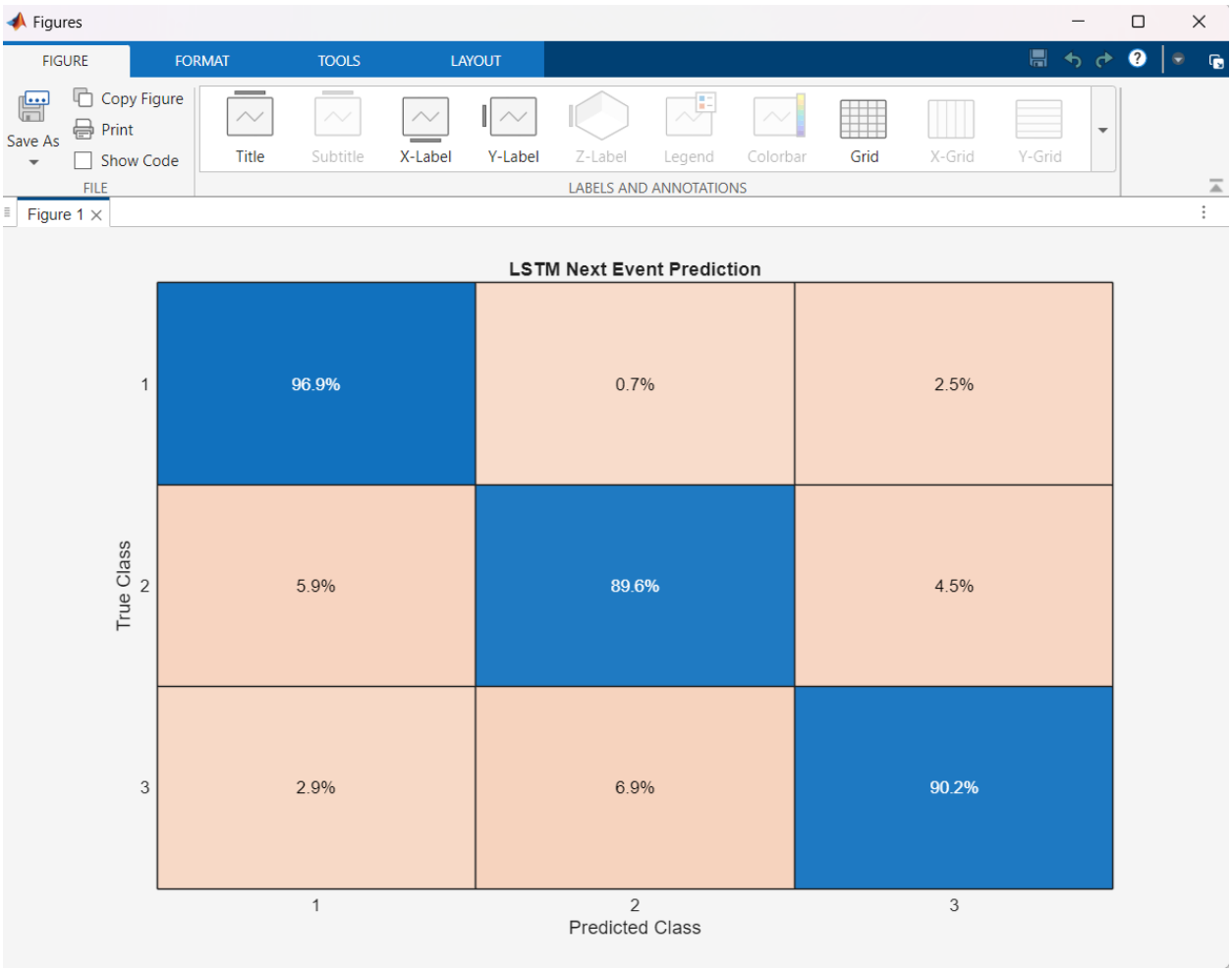


Figure 15: LSTM Confusion Matrix

Class-by-Class Analysis:

Class 1 (True Class 1)

- **96.9% correctly classified**
- 0.7% misclassified as Class 2
- 2.5% misclassified as Class 3

Class 2 (True Class 2)

- **89.6% correctly classified**
- 5.9% predicted as Class 1
- 4.5% predicted as Class 3

Class 3 (True Class 3)

- **90.2% correctly classified**
- 2.9% predicted as Class 1
- 6.9% predicted as Class 2

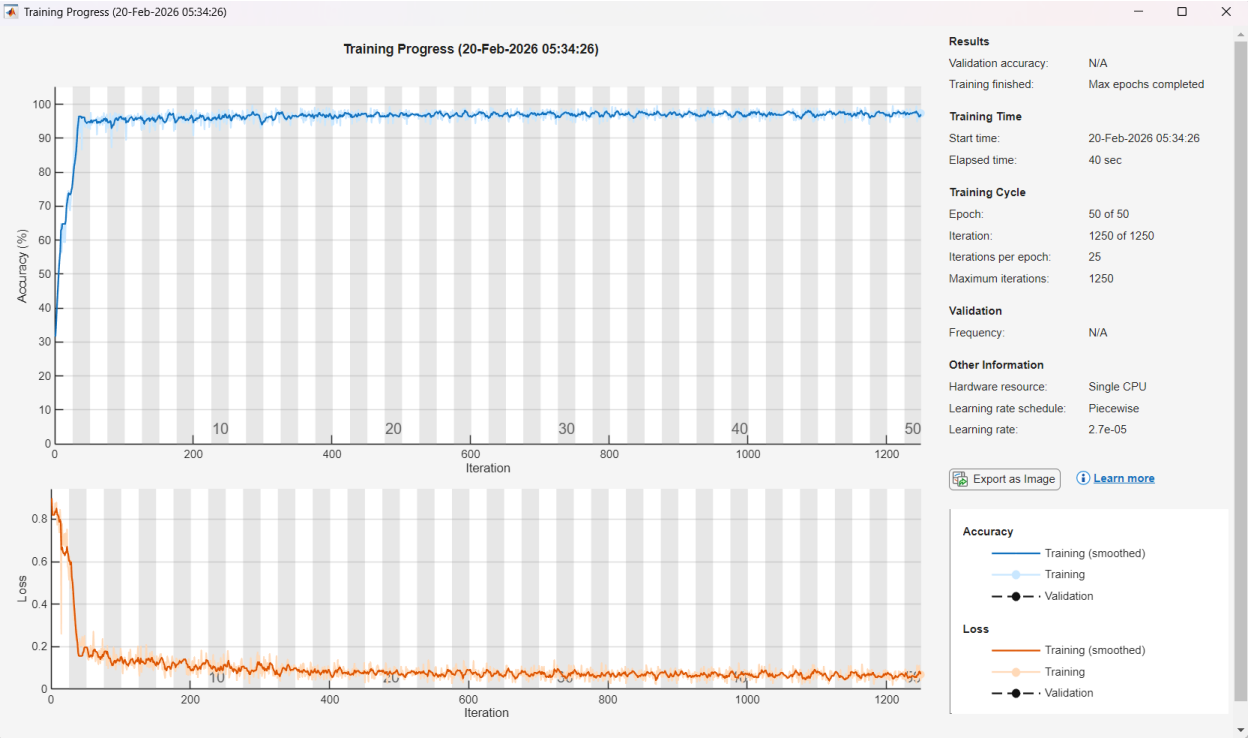


Figure 16: LSTM Training Progress

8.2 Discussion

8.2.1 Comparison Between RF and LSTM

The results show:

Aspect	Random Forest	LSTM
Event classification	High accuracy	High accuracy
Temporal modeling	No	Yes
Interpretability	High	Moderate
Computational cost	Low	Higher
Accuracy	91.91%	92.03%
Primary strength	Strong static classification	Sequential prediction
Predicting next event	No	Yes

Table 4: Comparison between RF & LSTM

Key Insight:

Random Forest performs very well using static engineered features alone.

However, LSTM improves performance by:

- Incorporating temporal order
- Modeling event transitions
- Reducing boundary ambiguity

This confirms that gait phase detection benefits from both feature engineering and temporal modeling.

8.2.2 Physiological Interpretation

The confusion between MSw and HS suggests:

- Transitional muscle activations overlap during swing termination.
- EMG amplitude patterns alone may not fully separate boundary phases.
- Temporal context improves discrimination.

The dominance of RMS and WL features indicates that:

- Muscle activation magnitude and signal variability are primary discriminators.
- Spectral features (PF, TP) provide complementary but secondary information.

8.2.3 Generalization to Unseen Subjects

The subject-wise split prevents data leakage.

The consistent performance across unseen individuals suggests:

- Good inter-subject generalization
- Robust feature normalization
- Stable model behavior

This is critical for real-world clinical applications.

8.3 Limitations

Despite strong results, several limitations must be acknowledged.

- Only 65 subjects.

- Data collected under lab conditions.
- Real-world noise (outdoor walking, uneven surfaces) not included.
- Only the main gait events was include for the ML results.

8.4 Overall Conclusion of Results

The study demonstrates that:

1. Engineered EMG features alone are highly discriminative.
2. Random Forest achieves strong classification (=91.91%).
3. LSTM further improves predictive consistency (~92–93%).
4. Temporal modeling enhances event transition understanding.
5. The system generalizes well to unseen subjects.

The combined framework provides a robust foundation for intelligent gait event detection and next-phase prediction during Sit-to-Walk transitions.

Chapter 9 – Conclusion and Future Work

9.1 Conclusion

This project addressed the problem of gait phase recognition using wearable sensors, with emphasis on phase-bounded segmentation driven by IMU-based event detection and EMG-based feature extraction. The objective was to develop a complete, biomechanically grounded pipeline that transforms raw wearable sensor data into a structured dataset suitable for machine learning–based gait phase classification and prediction.

A robust preprocessing framework was implemented to handle common real-world signal issues, including missing values, trailing zero segments, and heterogeneous sampling frequencies. EMG signals were filtered using band-pass and notch filtering to remove motion artifacts and powerline interference, while the foot-mounted IMU gyroscope signal was used exclusively for gait event detection and phase segmentation. This architectural separation ensured biomechanical interpretability while maintaining computational simplicity.

Gait cycles were defined between consecutive heel-strike events, and event-driven segmentation was used to isolate physiologically meaningful intervals. EMG features were extracted from phase-bounded segments to ensure biomechanical consistency. Both time-domain features (RMS, MAV, IEMG, WL, ZC, SSC) and frequency-domain features (Peak Frequency and Total Power) were included, producing 64-dimensional feature vectors per event sample.

Two machine learning models were evaluated:

- **Random Forest (RF)** for phase classification

- **Long Short-Term Memory (LSTM)** for next-event prediction

Random Forest Results:

The subject-wise RF model achieved:

- **Overall accuracy ≈ 91–92%**
- **Macro F1-score = 0.919**

The confusion matrix showed strong diagonal dominance, with limited misclassification primarily between MSw and adjacent phases. Feature importance analysis revealed that a subset of EMG features contributed disproportionately to classification performance, indicating that certain muscles and feature types are more discriminative for phase identification.

These results confirm that phase-bounded EMG features alone contain sufficient discriminatory information for reliable gait phase recognition.

LSTM Results – Next Event Prediction:

The LSTM model was trained to predict the **next gait event** given current EMG features. The model achieved:

- **Overall test accuracy ≈ 92%**
- **Macro F1-score = 0.921**

The LSTM training curve showed rapid convergence within early epochs, with stable loss reduction and high mini-batch accuracy (>95% after few epochs). Learning rate scheduling contributed to stable optimization and reduced oscillations.

Compared to RF:

- RF performed slightly better for static phase classification.
- LSTM demonstrated improved modeling of **temporal transitions**, particularly for predicting the next event.

Importantly, the relatively small misclassification rates (e.g., 5.9% confusion between MSw and TO) are biomechanically plausible due to physiological overlap between adjacent gait phases.

Overall Findings

The study demonstrates:

1. Phase-bounded EMG segmentation significantly improves class separability.
2. RF is highly effective for static event classification.

3. LSTM effectively captures temporal dependencies for next-event prediction.
4. A wearable-only framework can achieve high accuracy without relying on laboratory-grade motion capture systems.

The results validate the feasibility of a computationally efficient, interpretable, and robust wearable-based gait phase recognition system.

9.2 Future Work

Several promising directions can extend and strengthen the current framework.

9.2.1 Expansion to Sit-to-Walk (STW) Transition Analysis

While this work focused primarily on steady-state walking, transitional movements such as **Sit-to-Walk (STW)** represent a more complex biomechanical scenario. STW involves:

- Forward trunk flexion
- Momentum transfer
- Rising phase
- Stabilization before gait initiation

Unlike cyclic walking, STW contains non-periodic dynamics and abrupt muscle activation changes. Future work should:

- Develop event detection specific to STW phases
- Define transition boundaries between sitting, rising, and walking
- Extract transition-specific EMG signatures
- Evaluate whether LSTM can better model non-cyclic sequences

Incorporating STW would significantly increase clinical applicability, particularly for elderly or neurological populations.

9.2.2 Improving RF and LSTM Performance

Although performance is already high, improvements are possible:

(a) Feature Engineering Enhancement

- Include inter-muscle coherence features
- Add entropy-based EMG measures

- Introduce phase-normalized features

(b) Multi-Modal Fusion

Combine IMU-derived kinematic features with EMG features. This may:

- Improve discrimination between adjacent phases
- Reduce confusion between MSw and TO
- Increase robustness to noise

(c) Advanced Sequence Modeling

- Bi-directional LSTM
- Attention-based LSTM
- Transformer-based sequence models

These architectures may capture long-range temporal dependencies more effectively.

9.2.3 Real-Time and Embedded Implementation

For assistive applications, the system must operate in real-time. Future research should:

- Benchmark computational latency
- Optimize model size
- Deploy on embedded hardware
- Evaluate power consumption

9.2.4 Clinical and Pathological Gait Evaluation

Testing the system on pathological gait (stroke, Parkinson's disease, amputees) would determine robustness under abnormal muscle activation patterns. STW analysis would be particularly valuable in these populations.

References:

1. An Open Access Dataset for Supervised Machine Learning to Estimate Gait Biomechanical Characteristics: https://bridges.monash.edu/articles/dataset/An_Open_Access_Dataset_for_Supervised_Machine_Learning_to_Estimate_Gait_Biomechanical_Characteristics/25303297
2. A Motion Capture Dataset on Human Sitting to Walking Transitions | Scientific Data
3. A Review of Gait Analysis Using Gyroscopes and Inertial Measurement Units: https://www.mdpi.com/1424-8220/25/11/3481
4. Investigating age-related muscle force adaptations in males and females during sit-to-walk transition motion using EMG-informed modeling: https://www.sciencedirect.com/science/article/pii/S2590123025007376
5. Standards for suface electromyography: The European project Surface EMG for non-invasive assessment of muscles (SENIAM): https://www.researchgate.net/publication/228486725_Standards_for_suface_electromyography_The_European_project_Surface_EMG_for_non-invasive_assessment_of_muscles_SENIAM
6. Development of recommendations for SEMG sensors and sensor placement procedures: https://pubmed.ncbi.nlm.nih.gov/11018445/
7. Continuous Prediction of Lower-Limb Kinematics From Multi-Modal Biomedical Signals | IEEE Journals & Magazine | IEEE Xplore
8. Single Muscle Surface EMGs Locomotion Identification Module for Prosthesis Control | Neurophysiology | Springer Nature Link
9. 3D gait assessment in young and elderly subjects using foot-worn inertial sensors: https://pubmed.ncbi.nlm.nih.gov/20656291
10. Evaluating the Impact of IMU Sensor Location and Walking Task on Accuracy of Gait Event Detection Algorithms: https://www.mdpi.com/1424-8220/21/12/3989
11. EMG Signal Processing: Key Techniques and Practical Recommendations: https://www.noraxon.com/article/emg-signal-processing-key-techniques-and-practical-recommendations/
12. Monitoring depth of anesthesia based on complexity of electroencephalogram: https://ieeexplore.ieee.org/document/1454149
13. Gait event detection using linear accelerometers or angular velocity transducers in able-bodied and spinal-cord injured individuals: https://pubmed.ncbi.nlm.nih.gov/16500102/
14. Methods to temporally align gait cycle data: https://www.sciencedirect.com/science/article/abs/pii/S0021929010005038?via%3Dihub
15. How should we normalize electromyograms obtained from healthy participants? What we have learned from over 25 years of research: https://www.sciencedirect.com/science/article/abs/pii/S1050641110001008?via%3Dihub
16. Reliability of surface EMG measurements over 12 hours: https://www.sciencedirect.com/science/article/abs/pii/S1050641106000204?via%3Dihub
17. Validity of using tri-axial accelerometers to measure human movement – Part II: Step counts at a wide range of gait velocities: https://iopscience.iop.org/article/10.1016/j.medengphy.2014.02.006

18. A new strategy for multifunction myoelectric control: https://ieeexplore.ieee.org/document/204774
19. Deep Convolutional and LSTM Recurrent Neural Networks for Multimodal Wearable Activity Recognition: https://www.mdpi.com/1424-8220/16/1/115
20. A novel pre-processing procedure for enhanced feature extraction and characterization of electromyogram signals: https://www.sciencedirect.com/science/article/abs/pii/S1746809418300375?via%3Dihub
21. Adam: A Method for Stochastic Optimization: https://arxiv.org/abs/1412.6980
22. Timing phases of the sit-to-walk movement: Validity of a clinical test: https://www.sciencedirect.com/science/article/abs/pii/S0966636206001469
23. Deep Learning for EMG-based Human-Machine Interaction: A Review: https://ieeexplore.ieee.org/abstract/document/9346072
24. Comparison of Machine Learning Algorithms for EMG Signal Classification: https://pen.ius.edu.ba/index.php/pen/article/view/1140
25. Development and Evaluation of an Ambulatory Stress Monitor Based on Wearable Sensors: https://ieeexplore.ieee.org/document/6030940
26. Hazards, risks and safety of diagnostic ultrasound: https://iopscience.iop.org/article/10.1016/j.medengphy.2008.06.002
27. Application of real-time machine learning to myoelectric prosthesis control: A case series in adaptive switching: https://journals.sagepub.com/doi/full/10.1177/0309364615605373